

LABORATORY MANUAL/OBSERVATION

APPLIED PHYSICS LAB *(Common to All)*



I- B.Tech. (II semester)
2022-23

Prepared by
Physics Faculty

NAME: _____

ROLL.NO: _____

BRANCH: _____

VNR VIGNANA JYOTHI
INSTITUTE OF ENGINEERING AND TECHNOLOGY
(An Autonomous Institute under UGC)
Bachupally Post, Hyderabad-500 090

FOREWORD

Engineering education is inconceivable without laboratory instruction. In the laboratory, students can explore their understanding of the subjects being taught in the class room by placing their learning in context. Good laboratory and practical work shall greatly help the students develop expertise in critical enquiry, problem solving, experimental design, data analysis, presentation, and other important academic and professional abilities. Laboratory provides hands on experiences that form a key part in developing the understanding not only in the later years of an engineering program, but across the engineering profession which the student will later be embedded.

In general, it is expected to achieve three major objectives in the applied physics lab. First, the student learns how to be an experimenter. Second, the laboratory shall be a place for the student to learn new and developing subject matter. Third, laboratory courses help the student to gain deeper insight into the theoretical concepts and enable them to apply to real world problems.

The present Lab manual is prepared taking into consideration all these aspects and highlighting the importance of laboratory education. This manual distinguishes itself with notable key insights:

- Each experiment starts with objectives (that are intended to be learnt) and learning outcomes (the expertise gained after completion of the experiment).
- Principle and theory of the experiment were given as elaborately as possible.
- Key words were provided to make the student familiar with the concepts in technical terminology.
- At the end of the experiment questions were provided to celebrate the learning outcome through self-assessment.
- Schematic figures, circuit diagrams, model graphs and tabular forms were provided wherever necessary to make the student to easily capture the theme of the experiment and to record the measurements, observations, calculations and graphical representations.

Thus we hope and wish that the students understand the correlation between learning the theory in the class room and experimenting it in the laboratory after reading through the manual before going to the experiment. We are confident that this manual activates the students' practical learning.

With best wishes,

Physics Faculty

Applied Physics Lab @ VNRVJIET

As students step into Applied Physics Lab, they are required to focus their attention mainly on three aspects, the important initiatives envisaged for the benefit of the students' learning in the laboratory. They are: the **story board** and the **lab management board** (both are static presentations in form of posters) and the important **VNR Lab Protocol** (a dynamic presentation through multi media).

Story Board:

The objective of the Story Board is to introduce the wholeness of the lab learning in a nutshell. It is a visual display of the interconnectivity among all the experiments relating them to real world situations and of course, all that weaving around a beautiful story. Display of list of experiments, course objectives and course outcomes are provided to engage the students in the lab learning updating his/her knowledge and skills. The story and the narration are not static and at Applied Physics Lab, the outgoing students are encouraged to come up with innovative version of their own story with appropriate narration at the end of their lab sessions which greatly benefits the future students.

Lab Management:

This is yet another novel initiative in the laboratory process with a visual display board. Lab Management board lets the students plan to maximize the utilization of **time and resources** in the laboratory for their learning. The students are familiarized with the occupation and utility of the lab for various purposes, budget utilization in the lab, academic and research activities and floor plan etc. The material resources and the list of experiments prescribed in the syllabus are also displayed to make the students aware of the availability and limitations of materials and equipment and also enable them to map the required resources to a given experiment. Global practices like safety measures and safety equipment are also shown along with Dos and Don'ts (list of what a student shall do and shall not do).

VNR Lab Protocol:

The VNR Lab Protocol is a high level conceptualization of implementing the learning process in every lab through multi media. It is a dynamic presentation, with audio-visuals, that allows maintenance of laboratories to meet all the needs of the students; providing material and intellectual resources, linking experiments, presenting real world examples. Learning keywords and mapping them with concepts, designing the experiments, ethical reporting, time management and adhering strictly to the safety norms. VNR Lab Protocol provides an opportunity to the students to experience the learning in the lab as, "**Interconnected Whole rather than Isolated Contingent Parts**". That is it induces the **experience of holistic learning** in the laboratory. The student after engaging, involving, engrossing and accomplishing (the main four layer structure of the learning prescribed in the VNR Lab Protocol) in the Physics Lab realises '**Why is he/she learning what he/she is learning (WIL)**'.

Career Vision Approach For VNR Lab Protocol



“Engage” Layer
Whole Scenario



Career Vision
Real World Situations
Shadow Engineering

“Involve” Layer
Whole to parts



Lab resources-general
Lab introduction Protocol
Lego building blocs

Interconnected
Whole rather
Than Isolated
Contingency
Parts

“Indulge” Layer
Exercise &
Experiment



Research, Pool & Setup
Study & Appreciate
Dexterity & Skill
Part Achievement

“Accomplish”
Layer
Parts to whole



Correlate & Experience
Imagine & Create
Sum Total Achievement



•Career Vision Approach

•Success is Sum Total of Small Achievements

•7 Habits of Highly Effective People

The aims of the VNR Lab Protocol

1. To structure and conduct the lab course with '**Career Vision Approach**' and as continuous process against the present practice of conducting experiments in isolation, i.e., as "**Interconnected Whole rather than isolated contingent parts**".
2. To continuously map lab work on to class work and syllabus to enhance the depth of learning. In this process the student will understand, "**Why is he learning in the class what he is learning**".
3. To "**Engage**", "**Involve**" and "**Engross**" the students to make them more imaginative, creative and independent in problem solving through "**design and planning**" of experiments rather than just conducting experiments.
4. To "**Actively involve**" (not just physically present) the students in lab work by making them feel that they have "**accomplished**" a major project or task out of their experimentations in the lab supplementing current class room learning.
5. To impress and inculcate "**7 Habits of Highly Effective People**" by Stephen R. Covey as a part of lab procedures and Protocol. This training not only helps them to complete their experiments and projects successfully but also to become a good professional and a leader.
6. To make the students "**Industry Ready**" by transforming the lab into a "**Virtual Industry**".
7. To inculcate the abilities and values, traditional and classical lab "**Concepts and skills**" through intellectual and manual "**Dexterity**".
8. To define and devise metrics for students' skills with respect to "**placement and industry readiness**".

General Instructions

- All students must wear apron while in the lab
- Write down the tabular formats to enter all the measurements you are going to make.
- **Do not use pencil** for entering results in OBSERVATION book.
- Record several values of each measured quantity, to obtain accurate results, wherever possible.
- Write down the formulae to be used in the calculations.
- Show the calculations step by step.
- Finally enter the result with an estimation of error.
- Sketch the block or schematic circuit diagram, if any in your observation book.

A suitable scheme for recording the experiment is furnished below:

TITLE: Title of the experiment should be written on a fresh page with date clearly and neatly.

CIRCUIT DIAGRAM/ FIGURE: Diagrams or circuits should be drawn neatly on left side blank page.

PROCEDURE/METHOD FOR RECORDING THE MEASUREMENTS: Record all measurements meticulously. Enter all measurements in suitably drawn tabular format, giving the units.

CALCULATIONS: Convert all measured values into SI units where necessary and substitute them in the formulae. Show each and every calculation.

GRAPH: If a graph is plotted on graph sheet, choose the scales in such a way that the graph paper is maximum utilized. Label the two axes clearly with units. Use a sharp pencil to draw the curves.

RESULT OR CONCLUSION: Give the result to a sensible order of accuracy. It must state the units also.

Applied Physics Lab [L : 0; T:0; P :3 (1.5 credit)]

Course Objectives:

1. To understand the working principle of lasers and optical fibers.
2. To analyze the characteristics of semiconductor devices.
3. To measure the time constant of RC circuit and dielectric constant of material.
4. To study the behavior of magnetic materials and understand least square method.
5. To compare the experimental results with the classroom learning.

Course Outcomes:

After completion of the course the student is able to

1. demonstrate the total internal reflection in optical fiber using lasers.
2. realize importance of semiconductor devices and optoelectronics in daily life.
3. illustrate discharging of a capacitor and polarizability of dielectric material.
4. identify the importance of least square fitting and applications of magnetic materials.
5. correlate the experimental results with the classroom learning.

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Introduction to Instruments used in the Physics Lab

(i) VERNIER CALLIPERS

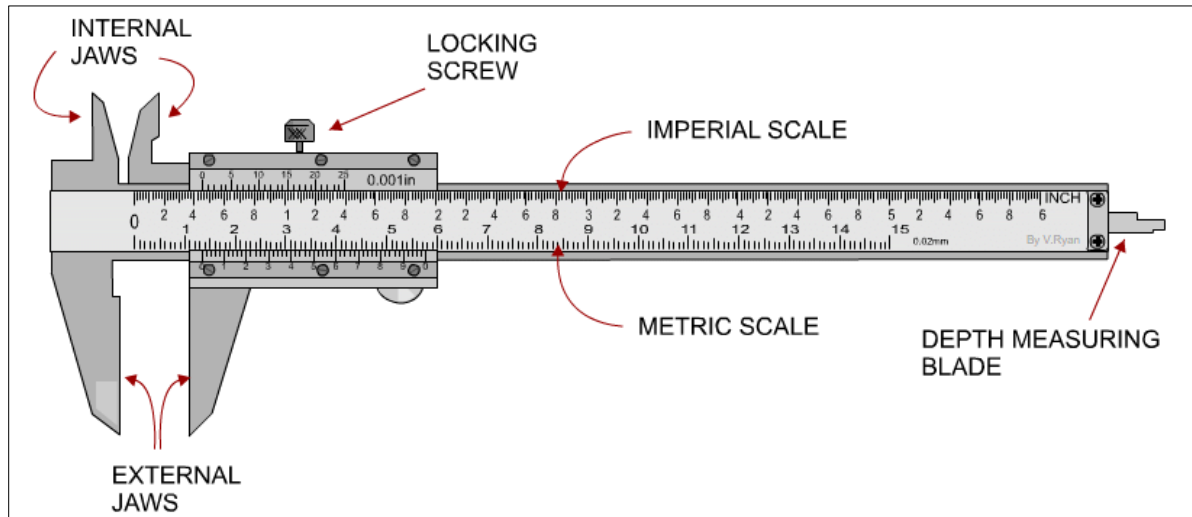


Fig i : Vernier Callipers

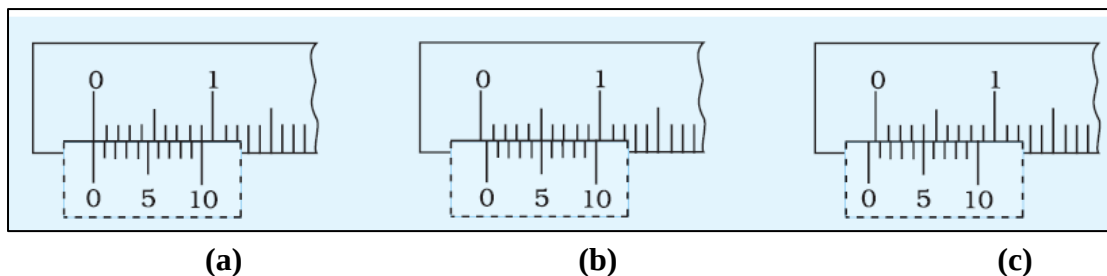


Fig.ii Zero error (a) no zero error (b) positive zero error (c) negative zero error

A vernier callipers is defined as a measuring device that is used for the measurement of linear dimensions. It is also used for the measurement of diameters of round objects with the help of the measuring jaws. French mathematician **Pierre Vernier** invented the vernier scale in 1631. The main use of the vernier calliper over the main scale is to get an accurate and precise measurement.

The least count of vernier calliper is also known as the vernier constant. It is defined as the difference between one main scale division (MSD) and one vernier scale division (VSD).

It is mathematically given as:

$$LC = 1 \text{ MSD} - 1 \text{ VSD} \text{ ----- (i)}$$

The principle of vernier calliper is

$$(N - 1) \text{ MSD} = N \text{ VSD, where } N = \text{number of vernier divisions}$$

$$N (1\text{MSD} - 1\text{VSD}) = 1 \text{ MSD}$$

$$N (LC) = 1 \text{ MSD}$$

$$LC = \frac{1\text{MSD}}{N} \text{ ----- (ii)}$$

Calculation of LC:

$$LC = \frac{1\text{MSD}}{N} \quad \text{where, } 1 \text{ MSD} = 1\text{mm and } N = 10$$

$$LC = \frac{1\text{mm}}{10} = 0.1 \text{ mm} = 0.01 \text{ cm.}$$

Thus, using vernier callipers we can measure length accurately up to 0.1 mm.

Table -1: Determining diameter of a cylinder

Sl.No.	Main Scale Reading M. S. R (cm)	Vernier coincidence V.C (divisions)	Diameter 'd' = M.S.R + (V.C x L.C) (cm)
1.			
2.			
3.			
Average diameter			d _{avg} =

Measuring the diameter of a small spherical or cylindrical body:

1. Keep the jaws of Vernier Callipers closed. Observe the zero mark of the main scale. It must perfectly coincide with that of the vernier scale. If this is not so, then take account of the zero error while using the instrument.
2. Look for the division on the vernier scale that coincides with a division of main scale. Position your eye directly over the division mark so as to avoid any parallax error.
3. Gently loosen the screw to release the movable jaw. Slide it enough to hold the sphere/cylindrical body gently (without any undue pressure) in between the lower jaws. The jaws should be perfectly perpendicular to the diameter of the body. Now, gently tighten the screw so as to clamp the instrument in this position to the body.
4. Carefully note the position of the zero mark of the vernier scale against the main scale. Usually, it will not perfectly coincide with any of the small divisions on the main scale. Record the **main scale division as MSR** just to the left of the zero mark of the vernier scale.
5. Start looking for exact coincidence of a vernier scale division with that of a main scale division in the vernier window from left end (zero) to the right. Note its number as **Vernier Coincidence (VC)** carefully.
6. Multiply 'VC' by least count (LC) of the instrument and add the product to the **MSR**. Ensure that the product is converted into proper units (usually cm) for addition to be valid. Thus, **total reading is MSR+ (VCxLC)**.
7. Repeat steps 3-6 to obtain the diameter of the body at different positions on its curved surface. Take three sets of reading in each case and record the readings in Table (i).
8. Enter the average diameter in result.

(ii) SCREW GAUGE:

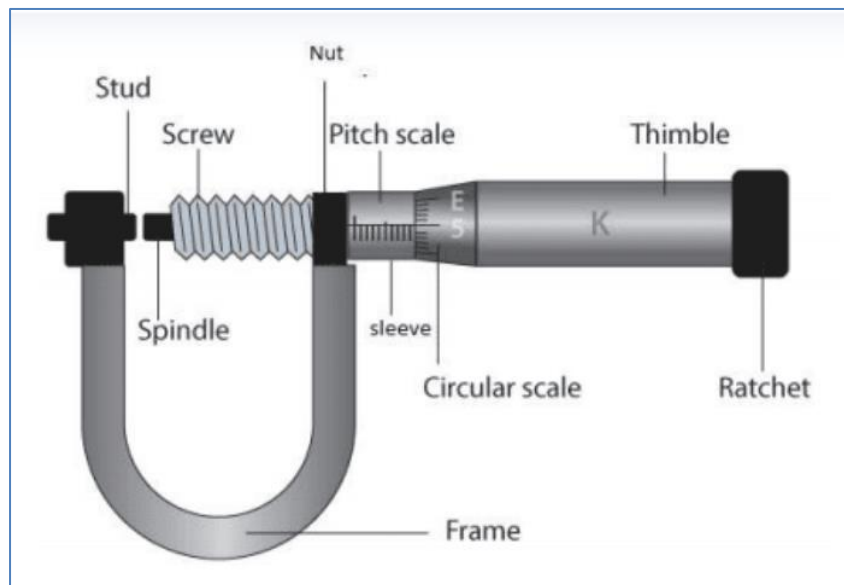


Fig iii: Screw Gauge

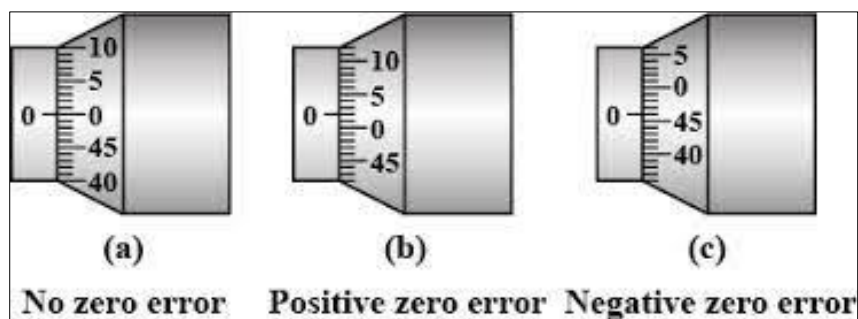


Fig iv: Error in Screw Gauge

Using vernier callipers we can measure length accurately up to 0.1 mm. To measure more accurately, up to 0.01 mm, we use screw gauge. In any ordinary screw, there are threads and the separation between any two consecutive threads is the same. The first screw gauge was created in the 17th century by *William Gascoigne*. Fig. ii shows a screw gauge.

Working principle: The distance advanced by the screw when it makes its one complete rotation is the separation between two consecutive threads. This distance is called the Pitch (p) of the screw or pitch scale division (PSD). It is usually 1 mm.

The Least count (LC) is the distance moved by the tip of the screw, when the screw is turned through 1 division of the circular. The least count can be calculated using the formula:

$$\text{Least count of the screw gauge} = \frac{\text{Pitch of the Screw}}{\text{no. of divisions on the Head scale}}$$

$$LC = \frac{1\text{PSD}}{N} \text{ ----- (iii)}$$

Calculation of LC:

$$LC = \frac{1\text{PSD}}{N} \quad \text{where, } 1 \text{ PSD} = 1\text{mm and } N = 100$$

$$LC = \frac{1\text{mm}}{100} = 0.01 \text{ mm} = 0.001 \text{ cm.}$$

Thus, using screw gauge we can measure length accurately up to 0.01 mm.

Table -2: Determining diameter of a wire
Error= **Correction=**

Sl.No.	Pitch Scale Reading (PSR) (cm)	Head Scale Coincidence (HSC) (divisions)	Corrected Head Scale Coincidence (CHSC) (divisions)	Diameter 'd' = PSR + (CHSCxL.C) (cm)
1.				
2.				
3.				
Average diameter				d _{avg} =

Measurement of diameter of a wire:

1. Bring the spindle in contact with the stud and calculate the zero error. If there is no zero error, then note down zero error nil.
2. Place the wire between spindle and stud. Rotate ratchet till the wire is fixed tightly and stop when it makes click sound.
3. Note the number of divisions of the pitch scale reading (P.S.R) visible before the edge of circular scale.
4. Note the number (N) of the division of the circular scale lying over reference line, which is known as Head Scale Coincidence (HSC). Apply zero correction and find Corrected Head Scale Coincidence (CHSC).
5. Multiply 'CHSC' by least count (LC) of the instrument and add the product to the **PSR**. Ensure that the product is converted into proper units (usually cm) for addition to be valid. Thus, **total reading is PSR+ (CHSCxLC)**.
6. Repeat steps 2, 3, 4 and 5 for three different positions separated equally throughout the length of the wire. And record the readings in Table (ii).
7. Take mean of different values of actual diameter.
8. Enter the average diameter in result.

(iii) BREADBOARD

A breadboard is used to build and test circuits quickly before finalizing any circuit design. The breadboard has many holes into which circuit components, both active elements like diodes, transistors, ICs and passive elements like resistors, capacitors, inductors etc can be inserted. A typical breadboard is shown below (bottom view with interconnections between different slots):

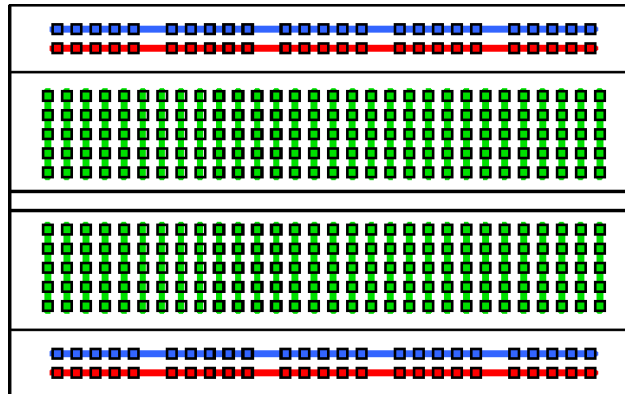


Fig v (a): Breadboard (Bottom View)

The bread board has strips of metal which run underneath the board and interconnect the holes on the top of the board. The metal strips are laid out as shown in the figure. Note that the top and bottom rows of holes are connected horizontally while the remaining holes are connected vertically. To use the bread board, the leads of components are placed in the holes. Each set of holes connected by a metal strip underneath forms a node. A node is a point in a circuit where two components are connected. Connections between different components are formed by putting their leads in a common node. The long top and bottom row of holes are usually used for power supply connections. The rest of the circuit is built by placing components and connecting them together with jumper wires. ICs are placed in the middle of the board so that half of the leads are on one side of the middle line and half on the other. A simple circuit might look like the following on a bread board.

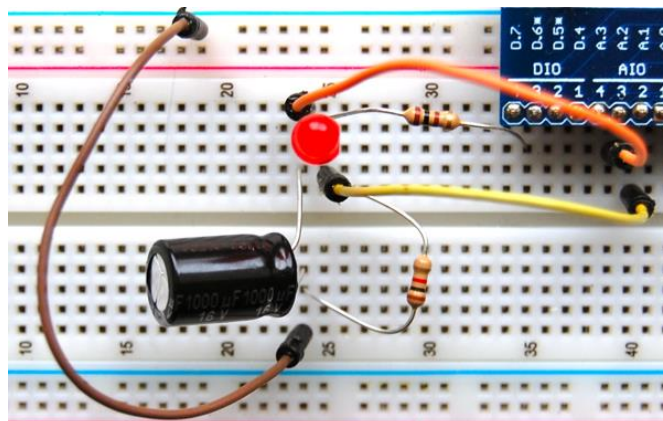


Fig v (b): Breadboard with connections (Top view)

To connect or test any circuit is important to place different components on the breadboard neatly and systematically. This makes the circuit function properly and the debugging (fault finding) also will be easier. It also helps when someone else needs to understand and inspect the circuit.

(iv) COLOR CODING IN RESISTORS:

Number	Color
0	Black
1	Brown
2	Red
3	Orange
4	Yellow
5	Green
6	Blue
7	Violet
8	Gray
9	White

Resistors:

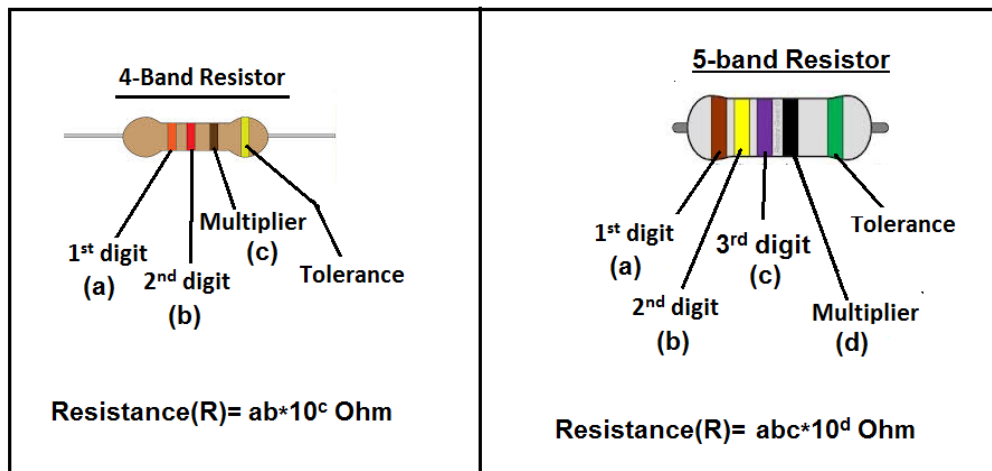


Fig iv : Resistor color codes for identification

The color that shows tolerance is either gold or silver. Gold and silver color shows the tolerance of 5 and 10 per cent of the prescribed values. Precision resistors are also manufactured with 1% tolerance for more sophisticated applications. The resistors are available for different power ratings, from 0.25 W to 10-20 W and more for different applications.

DATE:

Roll No.:

1. TORSIONAL PENDULUM: UNDERSTANDING THE METHOD OF LEAST SQUARES

Objectives:

- To determine the modulus of rigidity (η) of the material of the given wire by using Torsional Pendulum.
- To calculate the error in modulus of rigidity (η) in different methods.

Learning Outcomes: After completing this experiment the student is able to:

- Observe the rotational SHM with a small amplitude of rotation.
- Measure the time period (T) of oscillation.
- Recognize that T depends on the mass of the disc (M) the and length of the pendulum (l)
- Learn the graphical analysis of variation of length (l) with the time period (T)
- Differentiate the least squares method from conventional average value and graph methods in error analysis.

Resources required: A metal disc provided with a chuck nut at its center, metal wires, another chuck nut fixed to a wall bracket or a rigid clamp, a stopwatch, meter scale, screw gauge, vernier callipers, rough balance with weights, and graph sheets.

Keywords: Simple Harmonic Motion (SHM), angular acceleration, moment of inertia, torque, time period, rigidity modulus

Principle: For small twists, the restoring torque ' τ ' is proportional to the amount of twist ' θ ', the angular displacement (Hooke's Law). That is $\tau = -c\theta$.

Thus, the Torsional pendulum oscillates in Simple Harmonic Motion (SHM).

Formula: The modulus of rigidity, $\eta = \frac{4\pi MR^2}{a^4} \left(\frac{l}{T^2} \right)$ dyne/cm² (i)

where ' M ' is the mass of the disc, ' R ' is the radius of the disc,

' a ' is the radius of the wire, l is the length of the wire between chuck nuts,
and T is the time period of the pendulum.

$$\text{Error in } \eta = \left| \frac{\eta_{\text{actual}} - \eta_{\text{experimental}}}{\eta_{\text{actual}}} \right| \times 100 \% \text{ (ii)}$$

Experimental set-up:

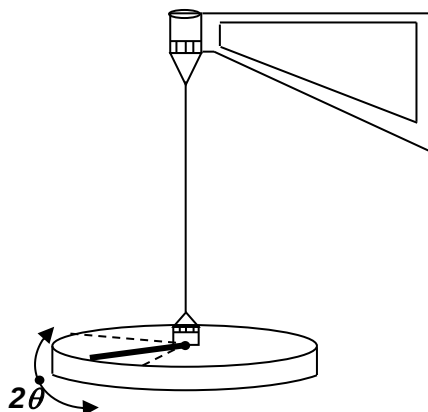


Fig.1: Torsional Pendulum

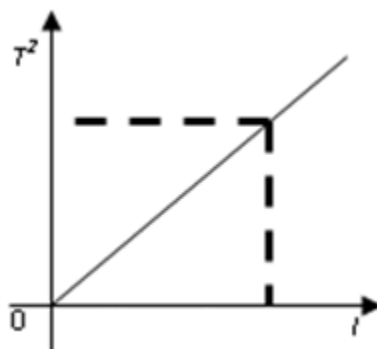
Graph:

Fig.2: Linear relation between I - T^2

Theory:

The Torsional pendulum consists of a uniform circular metal disc of about 8 to 10cm diameter with 1 or 2 cm thickness (different shapes and sizes can be considered provided moment of inertia can be determined easily), suspended by a wire at the centre of the disc as shown in figure1. The lower end of the wire is gripped into the chuck at the centre of the disc and the upper end is gripped into another chuck, which is fixed to a wall bracket.

When the disc is rotated in a horizontal plane, the wire gets twisted. The twisted wire will exert a torque on the disc tending to return it to the initial position. This is the restoring torque. For small twists, the restoring torque ' τ ' is proportional to the amount of twist ' θ ', the angular displacement (Hooke's Law). That is $\tau \propto \theta$; $\tau = -C\theta$ (1)

where C is called the Torsional constant and is defined as couple per unit twist of the wire. The negative sign shows that the torque is directed opposite to the angular displacement ' θ '. Equation (1) is the condition for angular simple harmonic motion.

The equation of motion for such a system is $\tau = I\alpha = I \frac{d^2\theta}{dt^2}$ (2)

where α is angular acceleration and I is moment of Inertia.

From Equation (1) and Equation (2) we get

$$-C\theta = I \frac{d^2\theta}{dt^2} \quad \text{or} \quad \frac{d^2\theta}{dt^2} + \omega^2\theta = 0$$

where $\omega = \sqrt{\frac{C}{I}} = \frac{2\pi}{T}$ is the angular frequency and T is time period.

Therefore, time period $T = 2\pi\sqrt{\frac{I}{C}}$ (3)

If ' a ' is the radius of the wire, ' l ' is the length of the wire between the chucks and ' η ' is the Rigidity modulus of the material of the wire, then the couple ' C ' per unit twist of the wire is given by

$$C = \frac{a^4\eta\pi}{2l} \quad \text{.....} \quad (4)$$

From Equation (3) and Equation (4) we obtain,

$$\eta = \frac{8\pi I}{a^4} \left(\frac{l}{T^2} \right) \quad \text{.....} \quad (5)$$

The moment of inertia, I , of a circular disc whose geometric axis coincides with the axis of rotation is given by,

$$I = \frac{MR^2}{2} \quad \dots\dots (6)$$

where ' M ' is the mass of the disc and ' R ' is the radius of the disc. Substituting the value of ' I ' from Equation (6) in Equation (5) we get,

$$\eta = \frac{4\pi MR^2}{a^4} \left(\frac{l}{T^2} \right) \text{ dyne/cm}^2 \quad \dots\dots\dots (7)$$

Procedure: First find the diameter of wire by using Screw gauge by taking the readings at three different points along the length (refer Table-I). Next the diameter of the circular disc is found out using Vernier callipers (refer Table-II). Mass of the disc is found out using common balance. The circular metal disc is suspended as shown in figure 1.

The length of the wire between the chucks is adjusted to 90cm. When the disc is in equilibrium position, a small mark is made on the disc to help to note the number of oscillations. The disc is set to oscillate by slowly turning it through a small angle. Care is taken to see that there are no lateral movements of the disc. The time for 20 oscillations is noted.

The procedure is repeated for the same length and the readings are noted as trial 2, and from the mean of the two recorded trials the time for 10 oscillations is calculated and from this, Time period of oscillation (T) is calculated (refer Table-III). The experiment is repeated, by decreasing length of the wire in steps of 5 or 10cm. Note to keep the length of the wire between 50 and 100 cm to get better results. The mean value of (l/T^2) is substituted in **Equation 7** and ' η ' is calculated.

A graph is also drawn with l on x-axis and T^2 on y-axis. It will be linear as shown in the **Figure 2**. From the graph the value of T^2 for the value of suitable length l is obtained and l/T^2 is calculated. This is substituted in **Equation 7** and ' η ' value is calculated. This is compared with the previous ' η ' value.

Table- I: To find diameter of the wire 'a' by Screw guage:

$$\text{Least count of the screw gauge} = \frac{\text{Pitch of the Screw}}{\text{no. of divisions on the Head scale}}$$

$$L.C. = \text{-----}$$

Zero error:

Correction:.....

Sl.No.	Pitch Scale Reading P.S.R (mm)	Head Scale Reading H.S.R (divisions)	Corrected Head Scale Reading C.H.S.R (divisions)	Diameter ' d ' = P.S.R + (C.H.S.R x L.C) (mm)
1.				
2.				
3.				

Average value of the diameter ' d ' = mm

Average value of the Radius ' a ' = $d/2$ = mm.

Table-II: To find radius of the disc 'R' by Vernier Callipers:

$$\text{Least count of the vernier calipers} = \frac{1 \text{ Main scale division value}}{\text{No. of Vernier Scale divisions}}$$

LC =-----

Sl.No.	Main Scale Reading (M.S.R) (cm)	Vernier coincidence (V.C) (divisions)	Diameter 'd' = M.S.R + (V.C x L.C) (cm)
1.			
2.			
3.			

Average value of the diameter ' d ' = cm

Average value of the Radius ' R ' = $d/2$ = cm.

Tale-III: To calculate l/T^2 :

Sl.No.	Length of the wire between chucks (l) (cm)	Time for 10 oscillations			Time period of oscillation ' $T = t/10$ ' (s)	T^2 (s ²)	l/T^2 (cm/s ²)
		Trial 1	Trial 2	Mean of Trails ' t '			
1.							
2.							
3.							
4.							
5.							

Average value of l/T^2 = (cm/s²)

Find (l/T^2) by Least Square Method:

Least-square method is the curve that best fits a set of observations with a minimum sum of squared residuals or errors. Let us assume that the given points of data are $(x_1, y_1), (x_2, y_2), (x_3, y_3), \dots, (x_n, y_n)$ in which all x 's are independent variables, while all y 's are dependent ones. This method is used to find a linear line of the form $y = mx + c$ as shown in Fig.3; where m is the slope, and c is the y -intercept.

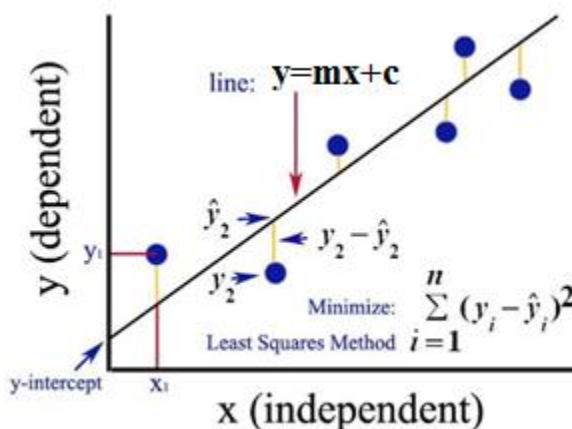


Fig.3: Fitting straight line by Least Squares Method

In the torsional pendulum experiment:

$$x = l \text{ and } y = T^2$$

$$\text{where, slope, } m = \frac{n \sum xy - \sum y \sum x}{[n \sum x^2 - (\sum x)^2]} = \frac{T^2}{l}$$

$$c = (\sum y - m \sum x)/n \approx 0$$

Here, n is the number of data points, substitute the value of m and b in the equation.
 $y = mx + c$; where $x = l$ (cm) and $y = T^2$ (s²)

Tale-IV: To calculate ' l/T^2 ' from Least Squares Method

S.No. (n)	x l (cm)	y T ² (s ²)	xy l x T ² (cm- s ²)	x ² l ² (cm ²)
1				
2				
3				
4				
5				
	$\sum x =$	$\sum y =$	$\sum xy =$	$\sum x^2 =$

$$\frac{T^2}{l} = m = \frac{n \sum xy - \sum y \sum x}{[n \sum x^2 - (\sum x)^2]} = \quad (\text{S}^2/\text{cm})$$

$$\Rightarrow \frac{l}{T^2} = \quad \text{cm/s}^2$$

Calculations:

☛ Actual value of rigidity modulus $\eta_{\text{actual}} =$ _____ dyne/cm²

☛ Radius of the wire 'a' = _____ cm

☛ Radius of the disc 'R' = _____ cm

☛ Mass of the disc 'M' = _____ g

☛ Average value of ' l/T^2 ' = _____ cm/s²

$$\frac{4\pi MR^2}{a^4} = \frac{4(3.14)(\quad)(\quad)^2}{(\quad)^4} \text{----- g/cm}^2$$

Substituting the value of ' l/T^2 ' from Table III

$$\eta_{\text{from observations}} = \frac{4\pi MR^2}{a^4} \left(\frac{l}{T^2} \right)$$

$$= (\quad)(\quad)$$

$$= \text{----- dyne/cm}^2$$

Substituting the value of ' l/T^2 ' from graph

$$\eta_{\text{from graph}} = \frac{4\pi MR^2}{a^4} \left(\frac{l}{T^2} \right)$$

$$= (\quad)(\quad)$$

$$= \text{----- dyne/cm}^2$$

Substituting the value of ' l/T^2 ' from least squares method

➤ $\eta_{\text{from least square method}} = \frac{4\pi MR^2}{a^4} \left(\frac{l}{T^2} \right)$

$$= (\quad)(\quad)$$

$$= \text{----- dyne/cm}^2$$

Safety Measures:

1. Radius of the wire 'a' is to be found accurately with the help of a screw gauge since it occurs as fourth power in the denominator.
2. Care should be taken that there are no horizontal oscillations of the disc.
3. The amplitude of the angular rotation should not be large.
4. The wire should be uniformly long, thin and straight without kinks.
5. The disc should be rotated through a small angle so that the twisting of the wire does not exceed the elastic limit.

Result: The rigidity modulus of the given wire is determined.

S.No.	Rigidity modulus of the given wire (η)	% of Error
1	η from observations =	
2	η from graph =	
3	η from least square method =	

Questions:

1. Does the rigidity modulus depend on the length and diameter of the wire?
2. On what factors does the time period depend?
3. Define the Time period of an oscillation?
4. Suppose, two wires of same material are given: one is long but thin and the other is short and thick. Which wire is preferred and why?
5. Why should we consider small angular amplitudes only?
6. What is S.H.M? Is Torsional pendulum an example of S.H.M?
7. How is Torsional pendulum different from simple pendulum?
8. What is rigidity modulus?
9. What is the dimensional formula for ' η '?
10. What is the shape of the curve if a graph is drawn between 'I' and 'T'? Where do the two curves, I vs T and I vs T^2 intersect each other?
11. What is method of least squares?
12. What is error? How do you determine it?
13. What are different types of errors? How do you eliminate them?

Demo:

<https://vlab.amrita.edu/index.php?sub=1&brch=280&sim=1518&cnt=4>

DATE:

Roll No.:

2. DETERMINE THE WIDTH OF GIVEN WIRE- USING LASER

Objectives:

- To obtain the Fraunhofer diffraction pattern due to a single slit or a thin wire using a laser source.
- To determine the wavelength of the source of light for known width of a given slit or vice-versa.

Learning Outcomes: After completion of the experiment the student is able to:

- Obtain a clear Fraunhofer spectrum on screen due to a single slit or a thin wire
- Determine the wavelength of the source of light
- Determine the slit width or thickness of the wire or hair
- Analyze qualitatively the changes that take place in the spectrum with variation of both the position and width of the slit.

Resources required: a single vertical slit or a thin wire or a strand of hair, laser light source, optical bench, meter scale and a screen (A-3 size white sheet).

Key Words: Superposition principle, diffraction, Huygene's principle, angular position, optical path difference etc.

Principle: When light passes through a vertical slit or a vertical object like a thin wire or hair, it undergoes diffraction (bending around the edges) to form alternate bright and dark fringes, provided the slit width or thickness of the wire is comparable to the wavelength of light incident.

Formula: Condition for maximum

$$a \sin \theta = (n+1/2) \lambda \quad \text{---- (i) } n = 1, 2, 3, \dots$$

where a = slit width or thickness of the wire, n = order of the spectrum, θ = angle of diffraction, λ = wave length of light

Theory: Diffraction is interference of a wave with itself. According to Huygene's principle waves propagate such that each point on a wave front acts as a new source. The sum of secondary waves emitted from all points in the wave front, propagate the wave forward. Interference between secondary waves emitted from different parts of the wave front can cause waves to bend around corners and cause intensity fluctuations much like interference patterns from separate sources.

The intensity distribution of monochromatic light diffracted from a single slit are based on

- i) Superposition principle
- ii) The wave nature of light, i.e., Amplitude of disturbance can be written as $A = A_0 \sin(\omega t + \phi)$ and intensity $I = (\sum A)^2$
- iii) Huygene's principle.

All the measurements and the estimations are based on the assumption that the distance L between the slit and the screen is much larger than the slit width ' a ', i.e., $L \gg a$. This is the condition to be satisfied for Fraunhofer diffraction to take place.

Single slit diffraction can be observed when light travels through a narrow vertical slit whose opening has a small width ' a '. Note must be taken that a similar diffraction pattern can be obtained if a thin wire of same width ' a ' is used instead of narrow slit. Light from different points across the width of slit will take different paths to arrive at a point on viewing screen. If the waves have same sign (that is in the same phase), then constructive interference takes place in which case the net amplitude is large and the light intensity is strong at that point. A bright fringe is obtained there. If they have opposite signs, however, they are out of phase, and destructive interference takes place where the net amplitude is small and the light intensity is weak. A relatively dark fringe is obtained at that point. It is these areas of strong and weak intensity distributions which make up the diffraction pattern on the screen. Note that the intensity is only redistributed, holding good the law of conservation of energy.

Let light of plane waves of wavelength λ incident on a single slit of width ' a ', properly aligned on an optical bench. The experimental set up is shown in the figure schematically. Let O be the position of the central bright maximum and let P be the position of first minimum on right side. Between O and P nearly midway the first order maximum is expected. Similar pattern is expected on the left side too. The pattern is symmetrical about the central maximum.

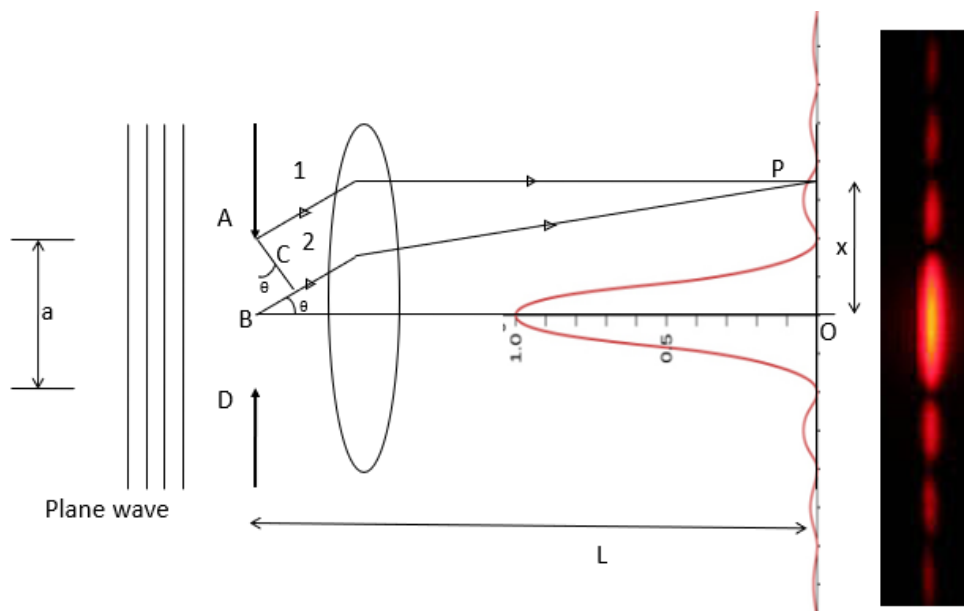


Fig 1: Plane wave incident on a single slit AD and the spectrum obtained on a screen kept at large distance L which is much greater than slit width ' a '

AD = slit width = a , and B is a point exactly located half way between A and D .
Hence $AB = BD = a/2$.

BC = path difference = $AB \sin \theta = (a/2) \sin \theta = \Delta$ ----- (1)

L = Distance between screen and slit (neglecting distance between slit and lens)

$OP = X$ = distance of first minimum from central maximum.

Assumption: $L \gg a$ (a condition for Fraunhofer diffraction)

The condition for minimum intensity at P ,

Δ = The path difference between rays 1 and 2 = $\lambda/2$,

Now point D in the slit is as far away from B ($=a/2$) as B is from point A. similarly for every point between A and B there is a corresponding point between B and D. Rays starting from any pair of corresponding points shall have the same path difference as the rays starting from A and B. Therefore general condition for **minima** at P due to superposition of rays originating from various points between A and D in the slit is given by (2) and (3)

$$\Delta = n\lambda/2. \text{ ----- (2) } n = 1, 2, 3, \dots$$

From figure $\Delta = (a/2) \sin\theta = n\lambda/2$

$$\Rightarrow a \sin\theta = n\lambda, n=1, 2, 3, \dots \text{ condition for minima ----- (3)}$$

The condition for maxima is not easy to obtain analytically, however for all practical purposes and calculations the condition is approximated to the relation (4).

$$a \sin\theta = (n+1/2) \lambda, \text{ with } n=1, 2, 3, \dots \text{ apart from the central maximum. ----- (4)}$$

Note: The condition for maxima is only approximated and not exact. Actually the maximum between any two successive minima lies not exactly in between but slightly towards the central maximum on either side.

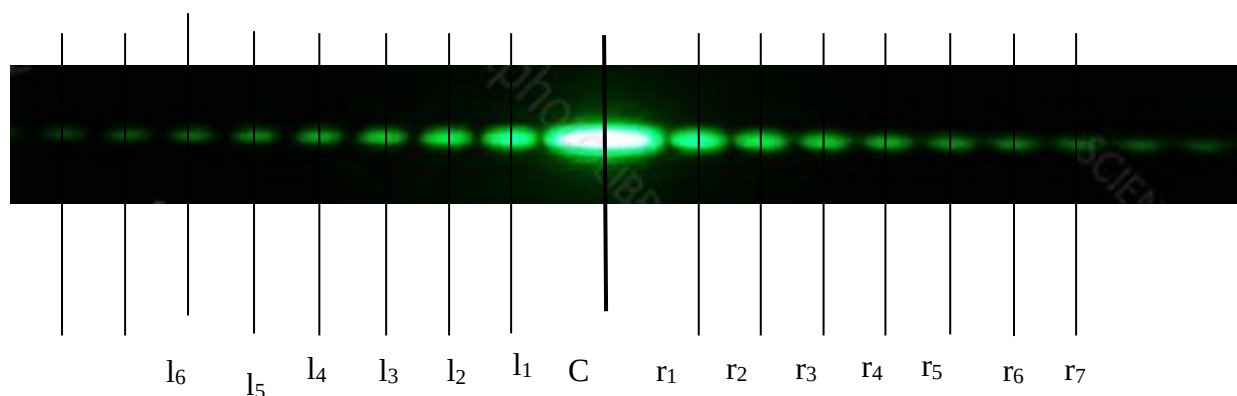


Fig.2: Pattern of the Single slit diffraction spectrum

Procedure: First fix the laser source, single slit with a very narrow opening on the optical bench. To find the thickness of a thin wire or strand of hair, the single slit can be replaced with the wire or hair. Keep a white sheet (preferably A3 size paper) as a screen near the wall. Keep the laser source far away from the slit and see that all of them (source, slit and the screen) are properly aligned. If the slit is not properly aligned the spectrum obtained will not be evenly distributed and distances of the given order from the central fringe on either side may differ.

Switch on the laser and a spectrum can be seen on the screen. Ensure that the distances of different orders from central maximum are same on either side respectively by adjusting the alignment. With a sharp pencil mark the bright spots on the white sheet carefully identifying the central bright spot. Measure the distance between the slit and screen and note it as L. Also note the distance between the source and the screen as D. Take out the sheet from the drawing board and draw parallel vertical lines passing through the pencil marks (bright spots) on the paper. Label the central bright spot as C and mark the lines as $l_1, l_2, l_3, \dots$ on left side and similarly $r_1, r_2, r_3, \dots$ on right side.

Measure the positions of $l_1, l_2, l_3, \dots$ and $r_1, r_2, r_3, \dots$. From the centre and tabulate the observations. The difference in the left and the right side readings for a particular order of the spectrum is twice the value of nth bright fringe separation from central bright fringe.

Substituting the values in the formula given above either the wavelength is measured (given the value of 'a') or the slit width 'a' or thickness of the wire or strand of hair is calculated (for given wavelength of laser source)

Observations:

D= Distance between Screen and Source=_____cm

L= Distance between Screen and Slit=_____cm

λ = Wavelength of given Laser source = _____A⁰

Table:

S.NO	Order of Diffraction (n)	Left side Readings l_n (cm)	Right side Readings r_n (cm)	$Y=(r_n+l_n)/2$	$\sin \theta = \frac{Y}{\sqrt{Y^2+L^2}}$	$a = \lambda (n+1/2) / \sin \theta$ cm
1						
2						
3						
4						
5						

Safety Measures:

1. Laser light is harmful to the eye. Avoid looking into the source directly.
2. Alignment of the source, slit and screen must be proper in order to get symmetrical spectrum about the central maximum.
3. Slit width should not be disturbed in the middle of the experiment.

Result: The width of the single slit/thickness of wire =cm

Questions:

1. What are the characteristics of light produced by a laser? Can this experiment be conducted by using a light source other than a laser?
2. What is the effect on the intensity distribution if the slit with 'a' is changed?
3. What would be the result if the experiment is to be carried out with white light?
4. What is the essential condition required to obtain a diffraction pattern?
5. What is the effect on the intensity pattern if the distance L between the slit the and screen is increased or decreased?
6. What is the effect on the intensity pattern if the distance (D-L) between the Source and the slit is increased or decreased?
7. If the size of the slit is as same the wavelength of the incident laser, what do you observe on the screen?

Demo: Variation of slit width, wavelength Vs Intensity, angle of diffraction

https://www.walter-fendt.de/html5/phen/singleslit_en.htm

DATE:

Roll No.:

3. DETERMINATION OF WAVELENGTH OF GIVEN LASER USING GRATING

Objectives:

- To obtain diffraction pattern due to multiple narrow parallel slits (Grating) using a laser source
- To determine the wavelength of the laser and angle of diffraction.

Learning Outcomes: After completion of this experiment the student is able to:

- Obtain a pure spectrum due to diffraction at multiple slits.
- Determine the wavelength of the light source.
- Relate the resolving power with number of lines in the grating.
- calculate the dispersive power of the grating and a width of principal maxima

Resources: Laser light source, grating, Blank Compact Disk, metal scale, screen (A3 sheet) retort stand, grating holder, optical bench.

Keywords: Superposition principle, coherence, diffraction grating, grating element, angular width, dispersive power, resolving power, principal maximum, interference, phase and path difference.

Principle: When plane waves are incident on an arrangement of equally spaced parallel narrow vertical slits, they undergo diffraction to form alternate bright and dark fringes.

Formulae:

$$1. \lambda = \frac{d \sin \theta}{m} = \frac{2.54 \sin \theta}{N \cdot m} \text{ cm}$$

$$2. \% \text{ error in } \lambda = \left| \frac{\lambda_a - \lambda_o}{\lambda_a} \right| \times 100$$

$$3. \theta_T = \sin^{-1} \left(\frac{m \lambda_a}{d} \right).$$

$$4. \% \text{ error in } \theta = \left| \frac{\theta_T - \theta_o}{\theta_T} \right| \times 100$$

where

θ = angular position of the fringe

m = order of spectrum

λ_a = wavelength of laser source (6500 Å)

d = slit width (grating element)

$$= \frac{1}{N} \text{ inches} = \frac{2.54}{N} \text{ cm}$$

N = number of lines (slits) in the grating

λ_o = Observed Value of Wavelength

θ_T = Theoretical angle of diffraction

θ_o = Observed angle of diffraction

Diagram:

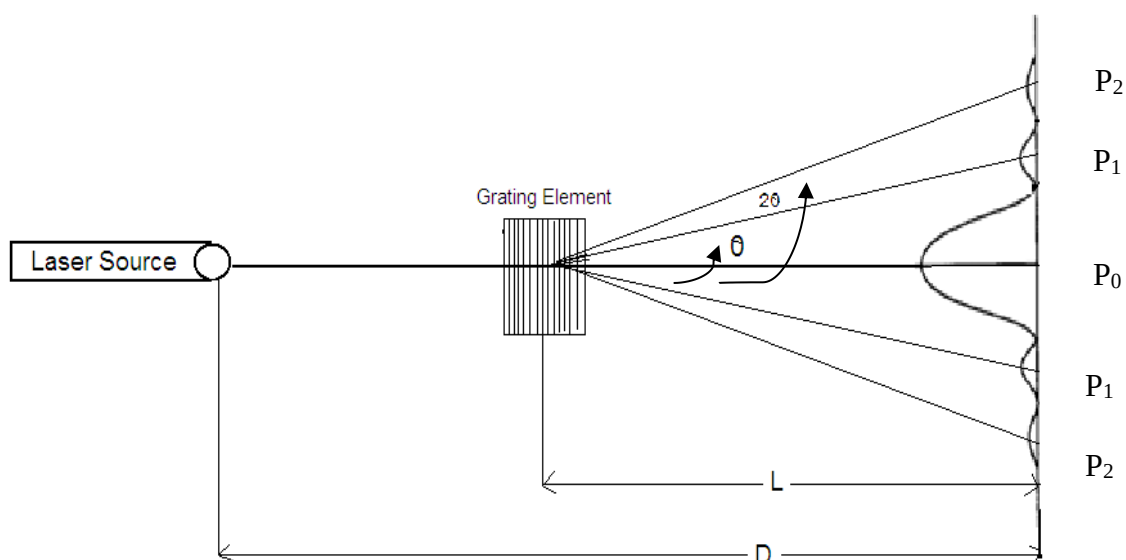


Fig 1: Schematic diagram for Fraunhofer diffraction due to multi slits (grating)

Theory: In a single slit diffraction pattern the central principal maximum (P_0) has most of the intensity and at the subsequent principal maxima ($P_1, P_2 \dots$) on either side the intensity falls off rapidly. The width of central bright envelop depends on the size of slit width 'a'. The minima on either side of the central maximum falls at angular position, θ_n determined by the equation

$$a \sin \theta_n = n\lambda, \text{ ---- (1)}$$

where ' λ ' is incident wavelength and ' n ' is order of the spectrum.

When an additional slit is provided, i.e in a double slit arrangement, the diffraction pattern due to individual slits is superposed with interference effects due to the two slits. The interference fringes have intensities that are determined by the intensity of the diffraction pattern at the location of a particular fringe. The width of the diffraction envelop depends on slit width and separation between interference fringes is governed by the slit separation.

Much the same is expected when the number of slits is increased. However, in case of multiple slits as the number of slits increased the bright fringes become narrower making it convenient to measure wavelengths more precisely. We consider here the interference patterns only for diffraction gratings, that is we assume we are near the central region of diffraction pattern.

A principal maximum occurs when the path difference between rays from any pair of adjacent slits, which is given by $d \sin \theta$, m is equal to an integral number of wavelengths, (Interference Effect)

$$d \sin \theta = m\lambda \text{ ----- (2)}$$

where $m=0, \pm 1, \pm 2, \dots$ is called order number, and d is slit separation.

A typical grating might contain large number of slits N distributed over an inch, equivalent to a grating spacing,

$$d = (1/N) \text{ inch} = (2.54/N) \times 10^4 \mu\text{m} \text{ (1 inch} = 2.54\text{cm} = 2.54 \times 10^4 \mu\text{m)}$$

Gratings are made by ruling equally spaced parallel grooves in thin layer of aluminium or gold deposited on a glass plate, using a diamond cutting point whose motion is automatically controlled by a ruling engine. Once such a master grating is prepared, replicas can be formed

by pouring a liquid plastic on the grating, allowing it to harden, and stripping it off. The stripped plastic fastened to a flat piece of glass or other backing form a good grating.

Some of the reflection types of gratings have different types of grooves on their surfaces. The angles of grooves are chosen so that light of a particular chosen order is reflected in a desired particular direction. In this the intensity of one particular order can be enhanced over that of other orders. Cutting the gratings in this way is called 'blazing' and most gratings in use today are blazed gratings.

The sharpening of principal maxima as N is increased is best understood from the expression obtained for angular width, $\Delta\theta$.

$$\Delta\theta = \lambda_a / (N.d.\cos\theta) \text{ ----- (3)}$$

This result gives the angular width for the principal maximum that occurs at an angle θ , corresponding to particular order, m . We can see that from the above expression

- (i) For $\theta=0$, $\Delta\theta = \Delta\theta_0 = \lambda / (N.d)$
- (ii) As the product Nd increases, $\Delta\theta$ becomes smaller.

The ability of a grating to produce spectra that permit precise measurement of wavelengths is determined by two intrinsic properties of the grating:

- The separation $d\theta$ between the spectral lines that differ in wavelength by a small amount $d\lambda$ and (2) the width or sharpness of the lines.
- The angular separation $\Delta\theta$ per unit wavelength interval $\Delta\lambda$ is called the dispersion D of the grating, or

$$D = d\theta / d\lambda = m / (d.\cos\theta) \text{ ----- (4)}$$

Another important feature of a grating is its ability to resolve closely spaced spectral lines of different wavelengths. Resolving power of a grating is defined as $R = \lambda / d\lambda$. Using Rayleigh's criterion for resolution and the expression for angular width $\Delta\theta$, it can be shown that

$$R = \lambda / d\lambda = Nm. \text{ ----- (5)}$$

Procedure:

Place grating parallel to the laser source such that the rays incident normally on the grating surface. Adjust the grating such that a clear diffraction pattern of the source is seen on screen or wall. Measure the distance of grating from wall let it be L and distance between the maxima (x). Take the readings of maxima on left and right side. Vary the grating distance from the screen and take the reading at different L . The corresponding x is to be measured. And tabulate the observed values and calculate wave length of source.

Observation and Tabular form:

Distance between Source and screen, $D = \text{_____ cm}$

Table-I: Determination of wavelength of laser source

Width of grating element: $N = \text{----- LPI}$; $d = 1/N = \text{----- cm}$

Order of maxima (m)	L cm	x cm			$\sin \theta = \frac{x}{\sqrt{x^2 + L^2}}$	θ (Degree)	$\lambda = (d \sin \theta) / m$ (cm)
		Left	Right	Mean			
1							
2							

Table-II: Determination of Error

Order (m)	% of error in λ	% of error in θ
1		
2		

Safety Measures:

1. Do not look into the Laser source directly.
2. Marking of the pattern should be taken carefully.
3. Alignment of the source, grating and screen is essential in order to obtain perfect symmetrical spacing between different spectral orders about the central maximum.

Result: The wavelength of the given LASER source and angle of diffraction is determined.

For $m=1$, $\lambda_1 = \text{----- } \text{\AA}$; and for $m=2$, $\lambda_2 = \text{----- } \text{\AA}$

For $m=1$, $\theta = \text{----- } ^\circ$; and for $m=2$, $\theta = \text{----- } ^\circ$

Questions:

1. What is grating element?
2. What is grating?
3. Are you using reflection type or transmission type grating?
4. Can you give example for transmission type of grating?
5. How does fringe separation depend on distance between the screen and the slit?
6. Are these fringes having equal widths?
7. Why central principal fringe is always bright?
8. To increase angle of diffraction for a given wavelength and order what should be done?

Demo: <http://ophysics.com/l5b.html>

DATE:

Roll No.:

4. DETERMINATION OF ACCEPTANCE ANGLE AND NUMERICAL APERTURE OF AN OPTICAL FIBER

Objectives:

- To determine the numerical aperture and acceptance angle of the optical fibers available.

Learning outcomes:

After completion of the experiment the student is able to

- Understand concept of light gathering capacity of optical fiber cable.
- Distinguishes the acceptance angle and acceptance cone.
- Realize phenomenon of total internal reflection

Resources Required:

Optical fiber cable, Fiber optic trainer module, N.A. measurement jig, NA Scale.

Keywords: total internal reflection, Numerical aperture, acceptance angle and acceptance cone.

Principle: Total internal reflection in an optical fiber

Formula:

Numerical aperture, $NA = W / (4L^2 + W^2)^{1/2}$ -----(i)

where, L = distance between end of fiber cable to screen
W = width of the spot.

And, $NA = \sin \theta_{\max}$

$$\theta_{\max} = \sin^{-1} (NA) \text{ -----(ii)}$$

where, $\theta_{\max}$ = acceptance angle,

Theory:

Numerical aperture of any optical system is a measure of how much light can be collected by the optical system. It is the product of the refractive index of the incident medium and the sine of the acceptance angle.

$NA = n_i \cdot \sin \theta_{\max}$; n_i for air is 1, hence $NA = \sin \theta_{\max}$

For a step-index fibre, as in the present case, the numerical aperture is given by $NA = (n_1^2 - n_2^2)^{1/2}$
For very small differences in refractive indices the equation reduces to $NA = n_1 (\Delta)^{1/2}$, where Δ is the fractional difference in refractive indices, n_1 and n_2 are refractive indices of core and cladding respectively.

Circuit diagram:

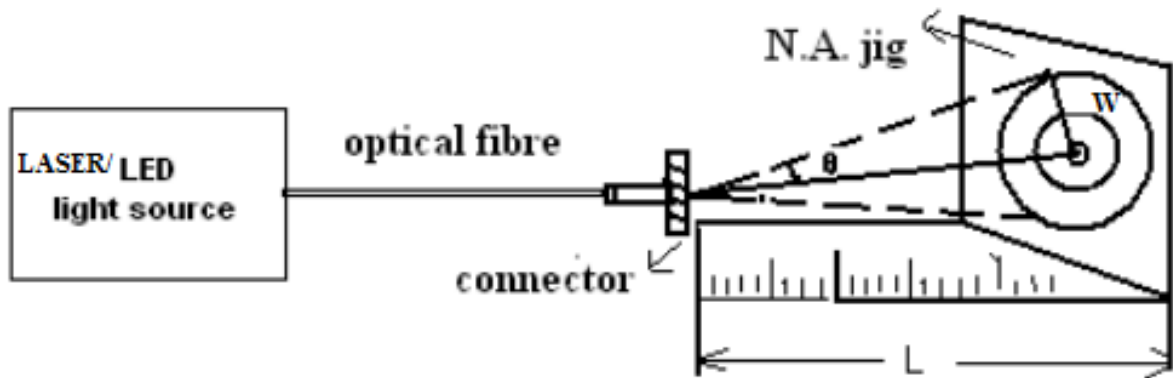
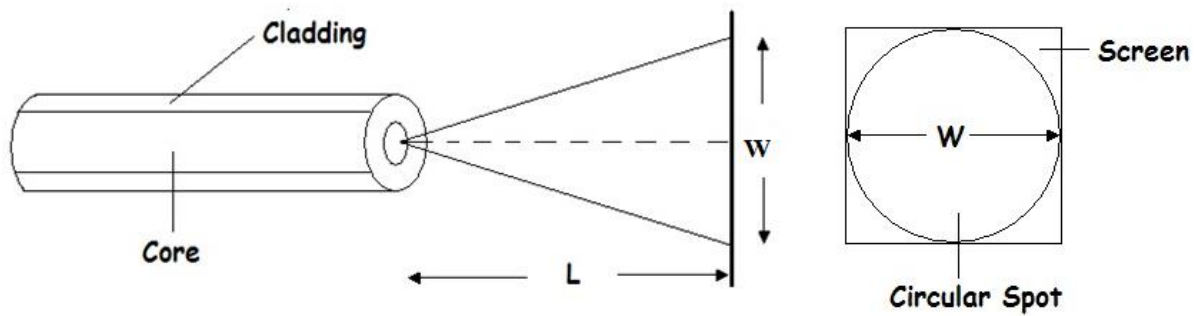


Fig: 1 Arrangement for NA measurement

Procedure:

One end of the optical fiber is connected to the N.A Jig through the connector, and the fiber end of the fiber is connected to the power out P of the N.A. module. The A.C mains is switched ON and the light passing through the cable at the other end of (coming to the N.A Jig) of the fiber is observed to ensure proper coupling is made or not. The “Set Po” knob is tuned in the clockwise direction to get maximum intensity of light through the fiber. The “set Po” is to be left free at this stage.

A screen with concentric circles of known diameter is kept vertically at a distance (L) from the fiber end and the red spot is seen on the screen.

The diameter (W) of the red spot is made exactly equal to the first concentric circle and the corresponding distance (L) from the fiber end to the screen is noted. The diameter of the red spot can be varied by the distance (L).

The experiment is repeated for the subsequent diameter of the circles by adjusting the length L. the diameter of the circle is determined using a travelling microscope.

For each set of observations, the numerical aperture is calculated using the formulae mentioned. Using the value of N.A formed, the acceptance angle can be calculated. The results are tabulated.

OBSERVATIONS & Table:

S. No.	Circle no.	L cm	W cm	NA	$\theta_{\max}$

Result: Numerical aperture and acceptance angle of the optical fiber is determined.

NA = ----- and $\theta_{\max}$ = -----

Questions:

1. Define Numerical Aperture?
2. What is critical angle?
3. What is the principle involved in OFC?
4. Define Total internal reflection?
5. Where do we see Total internal reflection in real world?
6. Tell two or three applications of OFC over other cables?
7. What are the units for NA?
8. What is acceptance angle?
9. What is acceptance cone?
10. Which type of light source you used in this experiment?
11. Why is it called optical fiber cable and with what material it is usually made up of?
12. What is the principle of TIR?
13. Can you describe real world situations of TIR?
14. Does dispersion has any effect in the cable?
15. What is launching angle and what is maximum possible launching angle?

DATE:

Roll No.:

5. DETERMINATION OF TIME CONSTANT OF AN RC CIRCUIT

Objectives:

- To study discharging characteristics of a capacitor through a resistance.
- To determine time constant of an RC circuit experimentally.

Learning Outcomes: After completion of the experiment, the student is able to

- Determines the value of resistor with the help of color code.
- Connect electric circuit by using bread board.
- Understand the exponential fall of charge in a capacitor
- Design an RC circuit with any desired time constant

Resources Required: Deflection Galvanometer, Capacitors and Resistors, bread board, Voltmeter (0-15V), Stop clock, D.C Source and connecting wires.

Key Words: capacitance, resistance, series and parallel combination, charging mode and discharging mode, exponential decay and time constant.

Principle: When a capacitor is connected to a source through a resistor, both the charging and discharging of the capacitor show exponential behaviour.

Formula: Theoretically, time constant $\tau = RC$ ----- (i)

where, R is resistance and C is capacitance in the circuit.

The current is proportional to deflection ,i.e, $I \propto d$, one can write $d = d_0 e^{-(t/RC)}$
when $t = RC$

$$d = d_0 \exp (-1)$$

$$= 0.365 d_0 \text{ ----- (ii)}$$

Circuit diagram:

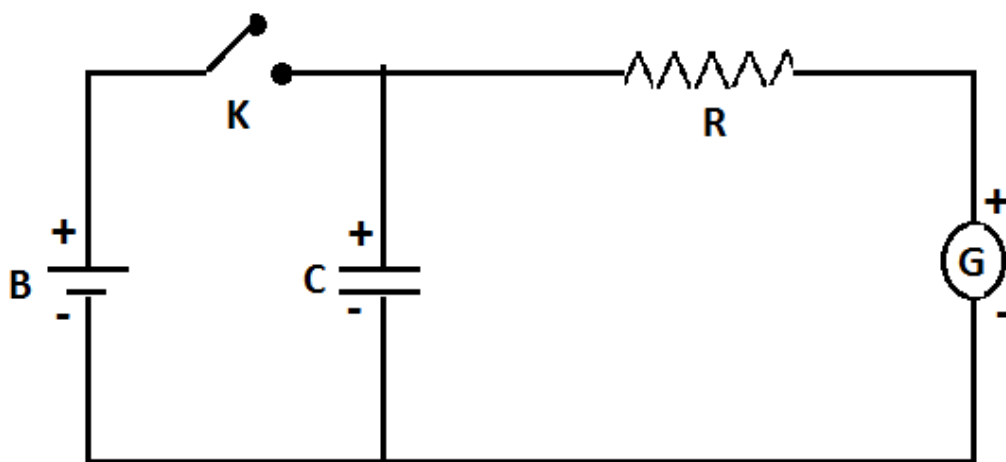


Fig1: Circuit diagram for discharging of capacitor

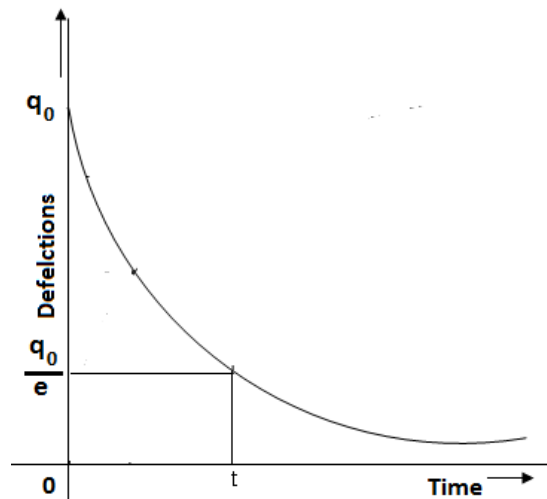


Fig 2: Discharging Deflections versus time curve for a capacitor through a resistor

Theory: The ability of a conductor to hold electric charge is called the capacity of a conductor. Different conductors at the same potential have different capacities for holding charge, depending on their size and shape. The capacity or capacitance of a conductor may be defined as the charge required raising its potential by unity. The unit of capacity is a Farad. Capacitor serves as devices for storing electrical energy. Capacitors are classified into different types depending upon the type of dielectric medium present between the plates, types of charges induced at the plates. Resistance is the characteristic of a particular specimen of the material. Its unit is a Ohm. Resistivity is the characteristic property of the nature of the material. In an electric circuit, when a capacitor and a resistor are connected to a source as shown in fig, electric charges build up across the capacitor. The discharge of the condenser takes place when the circuit is disconnected from the source of e.m.f.

When the tap key in circuit is pressed, a constant e.m.f works in the circuit, the condenser plates receive the charge till the potential difference across them becomes equal to E. When the key is released, the discharge of the condenser takes place. Let 'q' be the charge on the capacitor at the time 't' after the key is released. The instantaneous value of the p.d. across the capacitor is given by (q / c) and $E = 0$;

$$\text{i.e. } (q / c) - IR = 0 \text{ -----(1)}$$

$$\text{But } I = - dq / dt$$

$$\text{Therefore } (q / c) + R dq / dt = 0 \quad \text{or, } dq / dt = - q / RC$$

$$\text{Integrating } \log_e q = - (t / RC) + B \text{ ----- (2)}$$

Where B is the constant of integration.

At $t=0$, $q = q_0$; hence $B = \log_e q_0$

Substituting this value of B in equation (2), we get

$$\log_e (q / q_0) = -(t / RC) \text{ ----- (3)}$$

$$\text{Or } q = q_0 e^{-(t / RC)} \text{ -----(4)}$$

Therefore, Rate of discharge is

$$dq / dt = I = - q_0 (1 / RC) e^{-(t / RC)}$$

$$I = -I_0 e^{-(t/RC)}$$

Since current is proportional to deflection, i.e., $I \propto d$, one can write $d = d_0 e^{-(t/RC)}$

When $t = RC$

$$d = d_0 \exp(-1)$$

$$= 0.36 d_0$$

Thus, it is to be observed.

1. Smaller is the time constant, more rapid is the discharge of the capacitor.
2. The current in a R.C. circuit falls exponentially with time.

Now the fully charged capacitor is connected to a resistor in series with no emf in the circuit (see the circuit diagram for discharging). Let $t=0$, be the initial time when the switch is open and the charge on the capacitor plates be maximum, q_0 i.e., at $t=0$, $q=q_0$ (max). The instantaneous value of the p.d across the capacitor is given by q/c and across the resistor $-iR$, then by Kirchoff's laws Then one can write,

$$iR + q/C = 0 \dots\dots(7)$$

$$R dq/dt + q/C = 0 \dots (8).$$

Solving equation (8) for q , we get

$$q = q_0 e^{-t/RC} \dots\dots (9)$$

where q_0 is the initial charge on the capacitor. The current during discharge therefore

$$i = dq/dt = -(q_0/RC) \cdot e^{-t/RC} \dots\dots(10)$$

The negative sign indicates that the current in the circuit is in the opposite direction to the charging current. As the current in the circuit increases, the potential difference across the capacitor decreases from a maximum value $V_{\max,c} = q_0/C$. Therefore

$$V_c = q/C = (q_0/C) e^{-t/RC} = V_{\max,c} e^{-t/RC} \dots\dots\dots (11)$$

At $t = t_c$, the time constant,

$$q = q_0 e^{-t/RC} = q_0 / e = 0.37 q_0 \text{ (or)}$$

$$V_c = V_{\max,c} e^{-1} = 0.37 V_{\max,c} \dots\dots\dots(12).$$

Thus we can define time constant as the time of decay in which the charge decays to $1/e$ ($=0.37$) times the maximum, q_0 . Theoretically the charge becomes zero only at infinite time. However, the substantial decay takes place within the time interval equal to time constant.

Procedure: First sharpen the leads of the connecting wires and make the connections as per the circuit diagram. After making connections fix the resistance value in the Resistance box (RB) as given by the instructor. Switch on the circuit, the capacitor gets charged fully, as shown by the galvanometer maximum deflection. Now switch off the source. As the capacitor starts getting discharged the Galvanometer needle starts deflecting in reverse direction showing that the current decreases from maximum value towards zero. Start the stop clock at suitable initial deflection (given by the instructor) and take down the galvanometer readings at suitable intervals of time. Proceed until the deflected value falls off significantly below 0.36 times the

starting value. Repeat this for different values of resistance in R_B , and the observations are tabulated in Table 1. In each case, plot graph of Deflection (Y-axis) V_s Time (X-axis). After drawing a smooth curve note the point on Y-axis which is equivalent to $1/e$ times the maximum value on Y-axis. At this point draw a straight line parallel to X-axis until it touches the curve. From that point obtain the X-intercept which gives time constant graphically. Note this in Table II for comparison with theoretical values.

Observations:

Table I Discharging of capacitor in RC circuit

S. No.	Time (S)	Readings in the Galvanometer			
		$R_1 =$ $C_1 =$	$k\Omega$ μF	$R_{\text{effective}} =$ $C_{\text{effective}} =$	$k\Omega$ μF
1	0				
2	5				
3	10				
4	15				
5	20				
6	25				
7	30				
8	35				
9	40				
10	45				
11	50				
12	55				
13	60				
14	65				
15	70				
16	75				
17	80				
18	85				
19	90				
20	95				
21	100				
22	105				
23	110				
24	115				
25	120				
26	125				

Safety measures:

- The connecting wires should be cleaned well with sand paper
- Ensure all contact points in the circuit are properly established.
- The values of C and R should be selected in such a way that the rate of change of potential difference is gradual, that is neither too fast nor too slow
- Before switching on the power supply, ensure that the voltmeter reads zero
- The condenser should be completely discharged before charging it
- Until satisfied that all connections are proper do not switch on the mains

Results:

Table II

S.No.	Resistance k Ω	Capacitance μ F	Time constant Theoretical RC (s)	Time constant experimental from Graph (s)
1				
2				
3				

Questions:

1. When do you say a circuit is open?
2. What is a capacitor?
3. What is capacitance and what are its units in S.I system?
4. Does decaying of a capacitor time dependent? Explain graphically what is time constant of an RC circuit?
5. In the above circuit what is the use of S (the key) and what is the use of Galvanometer (G)?
6. Is it necessary G be connected in series always as shown? If so why?
7. What could be the resistance of the galvanometer?
8. What happens to time constant as resistance in the circuit increases?
9. If another similar capacitor is connected in parallel to the capacitor already existing in the circuit what happens to the time constant?
10. Is time constant characteristic of the capacitor used or is it the characteristic of the circuit itself?
11. What are the uses of a capacitor?
12. What is the physical significance of the time constant?

DATE:

Roll No.:

6. DETERMINATION OF ENERGY GAP OF A SEMICONDUCTOR

Objectives:

- To plot reverse biased diode current in logarithmic scale as a function of inverse temperature
- To determine the energy band gap of a semiconductor material using p-n junction diode.

Learning Outcomes: After completion of the experiment the student is able to

- Realize the temperature coefficient of resistance is negative in semiconductors.
- Observes that conductivity decreases logarithmically with inverse temperature.
- Shows that the energy band gap of the material is in the range of 1-2 eV as expected for a semiconductor material.
- Identify the importance of band gap of engineering materials particularly used in tunable opto-electronic devices like laser diodes, LEDs and photo diodes.

Resources Required: D.C. Power supply (0-15V), Junction diode (Germanium or Silicon), Thermometer (10°C – 100°C), heating (coil and oil bath) arrangement to heat the diode, Voltmeter (0-10V), Micro-ammeter (0-100 μ A).

Key Words: Semiconductor, Reverse bias, Energy band gap, Reverse saturation current

Principle: The conductivity of semiconductor increases with temperature.

Formula: Conductivity, $\sigma = C \exp(-E_g/2kT)$,(1)

where $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$, Boltzman Constant, E_g = energy band gap of the given semiconductor diode, T = absolute temperature($^{\circ}$ K)

Taking logarithms on both sides and in terms of current, equation (1) can be rewritten as $\ln(i) = \ln(c') - E_g/2kT$ (2)

When $\ln(i)$ Vs. $1/T$ is plotted, a straight line is obtained with a slope, $m = E_g / 2k$.

Therefore, $E_g = -2k (\text{slope})$ (J)..... (3)

Circuit diagram:

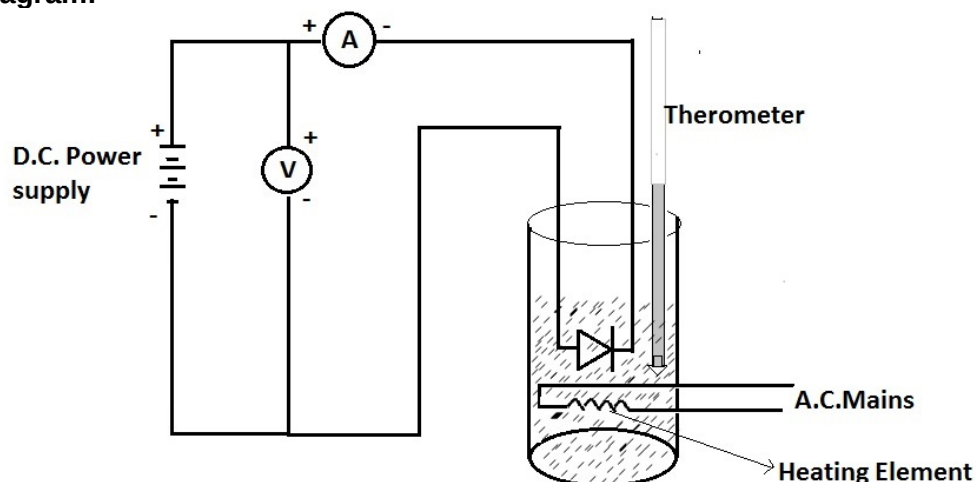


Fig. 1: Experimental set up for heating and cooling of a diode in reverse bias

Graph:

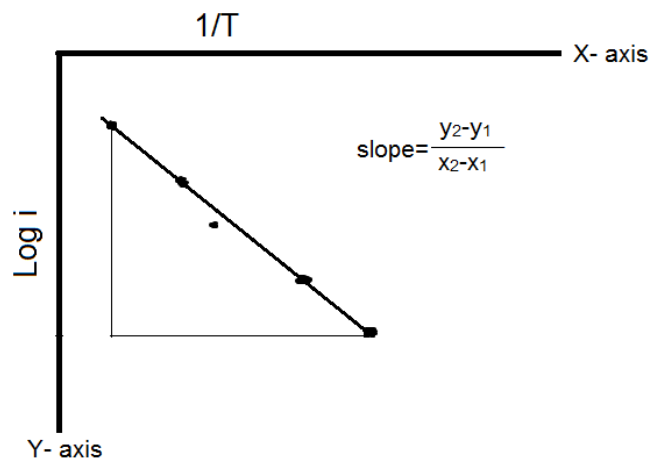


Fig. 2: Graph showing $\log i$ vs. $1/T$ (in fourth quadrant)

1. Draw a graph with $10^3 / T$ on X-axis and $\ln i$ on Y-axis.
2. The graph will be a straight line as shown in the fig.
3. Find the slope of the straight line, m from the graph.
4. Calculate the energy band gap of the given junction diode by substituting the Value of the slope (m) in equation.
5. Compare the calculated value of E_g , with the standard value.

Theory: The experimental arrangement comprises an oil bath which is provided with sockets at its mouth. The sockets are used to insert the thermometer and the semiconductor diode in the oil bath as shown in the figure. A heating element is fixed inside the oil bath which is used to raise the temperature of the oil bath by connecting to the A.C. main supply. The reverse biasing voltage can be adjusted by means of the voltmeter and the reverse saturation current can be measured with the help of a micro-ammeter.

PROCEDURE:

1. Identify the anode and cathode terminals of the given junction diode.
2. Connect the two terminals of the given junction diode (German or silicon) to the D.C power supply and micro ammeter in such a way that the diode is reverse biased.
3. Immerse the diode in the oil bath.
4. Insert the thermometer in the oil bath at the same level as that of the diode.
5. Switch ON the D.C. power supply and adjust the reverse bias voltage to 5 volt.
6. Switch ON the A.C. main supply, then the temperature of the oil bath gradually increases. Consequently, the current through the diode also increases.
7. When the temperature of the oil bath reaches to about 65°C , switch OFF the A.C supply. Stir the oil by means of a stirrer.
8. Consequently, the temperature of the oil bath will rise and stabilizes at about 70°C .
9. Note down the temperature of the oil bath and the current through the diode.
10. After few minutes, the temperature of the oil bath will begin to fall and the Current through the diode decreases.
11. Note down the value of the current for every 5°C decrease of the temperature, until the temperature of the oil bath falls to the room temperature.
12. Note down the observations in table.

Note:

1. When the Germanium diode is employed in the experiment, the temperature should not be increased beyond 80°C. This is because at higher temperatures, the Fermi level moves towards the centre of the forbidden energy gap and the junction properties are destroyed.
2. When silicon diode is employed, then the temperature should not exceed 200°C.
3. This experiment can also be performed for different reverse biasing voltages.

Table:

Cooling					
S.No	Temperature		Current (μA)	1/T (K^{-1})	$\text{Log}_e i$
	t ($^{\circ}\text{C}$)	T=t+273 (K)			
1	65				
2	60				
3	55				
4	50				
5	45				
6	40				
7	35				
8	30				

OBSERVATIONS:Room temperature = ----- $^{\circ}\text{C}$

Voltage: _____ (v)

 $Y_1 =$ $Y_2 =$ $X_1 =$ K^{-1} $X_2 =$ K^{-1} **Calculations :** Slope = $(Y_2 - Y_1)/(X_2 - X_1) =$ ----- K $E_g = -2k \text{ slope (J)}$ where $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$ $= -2 (\quad) (\quad) \text{ J}$ $=$ ----- J $1 \text{ eV} = 1.6 \times 10^{-19} \text{ Joules}, \Rightarrow 1 \text{ Joule} = \frac{1 \text{ eV}}{1.6 \times 10^{-19}}$ $E_g = \frac{(\quad)}{(\quad)} \text{ eV}$ $E_g =$ ----- eV

Safety measures:

1. The diode and the thermometer should be immersed at the same level in the oil bath.
2. The temperature and the current should be noted simultaneously.
3. When the experiment is performed temperature should not exceed 65°C.
4. This experiment should be performed by connecting the reverse biased position.

RESULT:

Energy gap of given semiconductor material is, $E_g = \text{-----} \text{ (eV)}$

QUESTIONS:

1. Define energy gap?
2. Define electron-volt?
3. What is the advantage of taking readings while cooling?
4. Is the diode connected in forward bias or reverse bias? Why?
5. Why to take oil? Why not water?
6. Why is the slope negative?
7. What is the relation between intrinsic charge carrier concentration and energy gap?
8. What are direct and indirect band gap semiconductors and give examples?
9. What is negative temperature coefficient of resistance? Give an example.
10. What is positive temperature coefficient of resistance? Give an example.

DATE:

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7. V-I CHARACTERISTICS OF LIGHT EMITTING DIODE (LED)

Objectives:

- To plot the V-I characteristics of Light Emitting Diode (LED) of a given color.
- To find the threshold potential for individual colors.
- To determine Energy band gap and hence, cut-off wavelength.

Learning Outcomes: After completion of this experiment, the student is able to

- Realize that a threshold potential or cut-off voltage is necessary for given LED to emit light.
- Show that the threshold voltage is different for different color LEDs, emitting light at different wavelengths.
- Reason out that the color of LED depends on band gap energy of semiconductor material.
- Observes that the cut-off wavelength is inversely proportional to the energy band gap.

Resources Required: LED characteristics kit, Ammeter, Voltmeter and connecting wires.

Keywords: Electroluminescence, light emitting diode, Energy gap, cut-off wavelength, recombination, threshold voltage, forward bias.

Principle: Recombination of electrons and holes in a properly doped diode in forward bias emit incoherent photons in visible region (Electroluminescence).

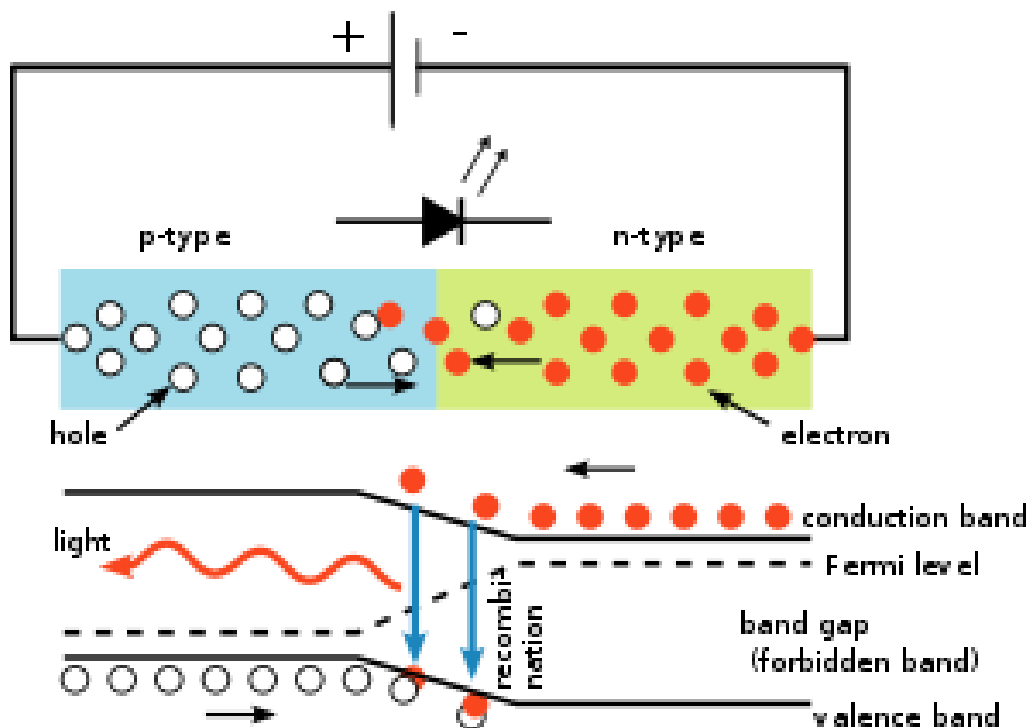


Fig. 1: Schematic diagram of recombination of electrons and holes in forward biased LED.

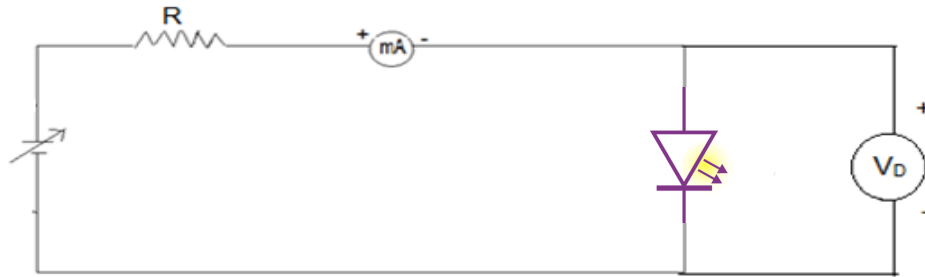
Circuit Diagram:

Fig. 2: Circuit diagram for studying V-I characteristics of LED.

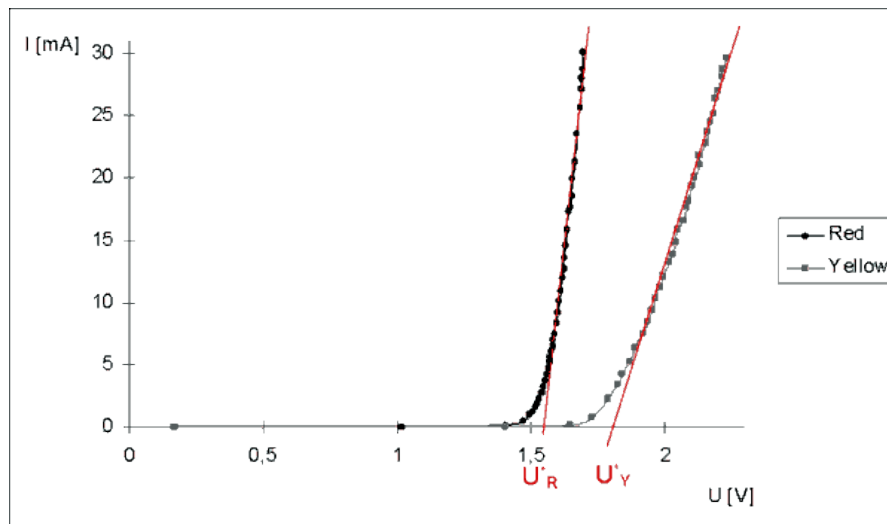
Graph:

Fig. 3: V-I Characteristics of Red and Yellow LEDs.

Theory: The Light Emitting Diode (LED) is basically a forward biased PN junction diode which can emit incoherent light or which can provide electro luminescence when it is energized. The LED emit only when it is forward biased and in reverse biased condition it acts like an ordinary reverse biased diode. More number of charge carriers will cross the junction because of reduced barrier height in forward bias, then some of them will recombine with the charge carriers of opposite polarity. As electron-hole pair recombines resulting in the release of excess energy through an emission of electromagnetic radiation. Thus LED emits light radiation as it is forward biased. The voltage at which the LED starts emission is called “cut-off voltage”. Recombination of electron-hole pairs can take place via two methods: (i) Direct recombination takes place in direct band gap materials like for example compound semiconductors such as GaAs, GaP, GaAsP, SiC etc. (ii) Indirect recombination takes place in indirect band gap materials of those elemental semiconductors like Si, Ge etc. Emission of light in LEDs, which is incoherent, happens in direct band gap materials only.

In direct band gap materials, the minimum energy of the conduction band lies directly above the maximum energy of the valence band in momentum versus energy plot. In this material, free electrons at the bottom of the conduction band can recombine directly with free holes at the top of the valence band, as the momentum of the two particles is the same. This transition from conduction band to valence band involves photon emission (takes care of the principle of energy conservation). This is known as direct recombination. Direct recombination occurs spontaneously and GaAs is an example of a direct band-gap material.

The wavelength of the light emitted, and hence the color, depends on the band gap energy of the materials forming the p-n junction. The emitted photon energy is approximately equal to the band gap energy of the semiconductor. The following equation relates the wavelength of the light emitted and the energy band gap.

$$h\nu = E_g = hc/\lambda \text{ ----- (1)}$$

$$\text{Therefore, } \lambda = hc/E_g \text{ ----- (2)}$$

Where h is Planck's constant, c is the speed of the light and E_g is the energy band gap. Thus, a semiconductor with a 2 eV band-gap emits light at about 620 nm in the red region; and a 3 eV band-gap material would emit at 414 nm in the violet region of the electromagnetic spectrum.

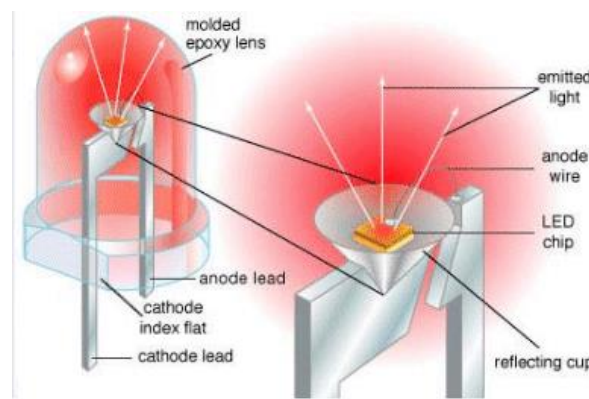


Fig. 4: Depicting the LED structure

The LEDs realized using two differently doped semiconductors that are the same material is called a homo-junction. When they are realized using different band gap materials they are called a hetero-structure device. The light emitted at hetero-structure LED is brighter than a homo-Junction LED. LEDs are usually built on an n-type substrate, with an electrode attached to the p-type layer deposited on its surface. P-type substrates, while less common, occur as well. Many commercial LEDs, especially GaN/InGaN, also use sapphire substrate.

Table: Typical values of energy band gap and threshold voltage of certain materials.

Color	Material	Band Gap (eV)	Threshold frequency (ν_t)	Threshold Wavelength (nm)
Blue	SiC	2.64	3.2 – 4.9	460
Green	GaP	2.19	2.2 – 2.5	520
Yellow	GaP ₈₅ As ₁₅	2.11	2.1 – 2.5	570
Orange	GaP ₆₅ As ₃₅	2.03	1.9 – 2.2	610
Red	GaP ₄₀ As ₆₀	1.91	1.7 – 2.7	660

TABULARFORM:

The voltage of the source is slowly increased. Observe the voltage at which the LED starts to glow. This is minimum voltage below which the LED cannot emit radiation.

Then a graph is drawn with voltage on x-axis and Current on y-axis. The curves are as shown in the graph.

For a given color, the curve is non-linear. Initially the current response is slow and then rapidly rises. A straight line drawn through the rapidly rising points intercepts the x-axis at cut-off voltage.

Safety Measures:

1. Connect the circuit properly.
2. Apply the required voltages where needed.
3. Do not apply physical stress on components.

Result:

1. V-I characteristics for a given LEDs are Plotted.
2. The cut off voltage for _____ color LED at resistance _____ ohms is _____ V.
3. The cut off voltage for _____ color LED at resistance _____ ohms is _____ V.

QUESTIONS:

1. Expand LED?
2. What is the working principle of LED?
3. What are the applications of LED?
4. On what factors the output color of LED depend?
5. Define cut-off voltage?
6. Does the resistance affect the cut-off voltage of LED?
7. What is a diode?
8. Does LED emit in forward or reverse bias?
9. Why are we drawing arrow on battery symbol?
10. What are O-LEDs?
11. Show that $E_g = 1242/\lambda$; where E_g is in terms of eV and λ is in terms nm.

DATE:

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8. V-I CHARACTERISTICS OF SOLAR CELL

Objectives:

- To obtain the V-I characteristics of a given solar cell.
- To determine fill factor of the solar cell.

Learning Outcomes: After completion of this experiment, the student is able to

- Understand the concepts of photo-electric effect, pair generation and transport of charge carriers in semiconductors
- Realize working of solar panels for power generation, which indeed useful as automatic street lights, solar cookers, calculators, photo-resistors etc.

Resources Required: Trainer board, solar cell, source of light, voltmeter (0-10v), ammeter (0-15 mA), Rheostat and connecting wires

Keywords: Photo-voltaic effect, reverse bias, photodiode or solar cell, fill factor and series resistance.

Principle: Conversion solar energy into electricity by using semiconductor panels exposed to the sun light, which is known as Photo-voltaic effect.

Formula: Fill factor of the cell, $FF = [P_m / I_{sc} V_{oc}]$ (i)

Where, P_m = maximum power point, V_{oc} = open circuit voltage and I_{sc} = short circuited current.

Series resistance of the solar cell: $R_s = [\Delta V / \Delta I]$ (ii)

Where, ΔV = change of maximum voltage; ΔI = change of maximum current with source.

Circuit Diagram:

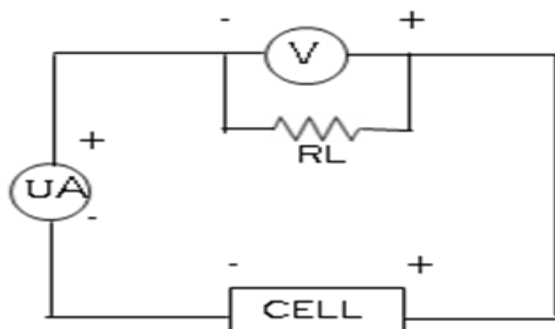


Fig. 1: Circuit diagram of solar cell

Graph: Current is taken on Y-axis and voltage is taken on X-axis.

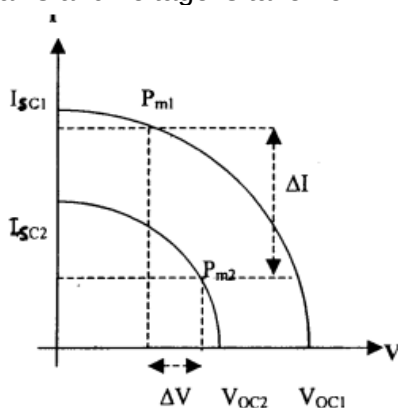


Fig. 2: V-I Characteristics of solar cell

Theory: Solar cell is basically a two terminal p-n junction device designed to absorb photon absorption through the electrical signal or power in the external circuits. Therefore it is necessary to discuss the physics of semiconductor p-n junction diode, which converts the optical energy into electrical signals.

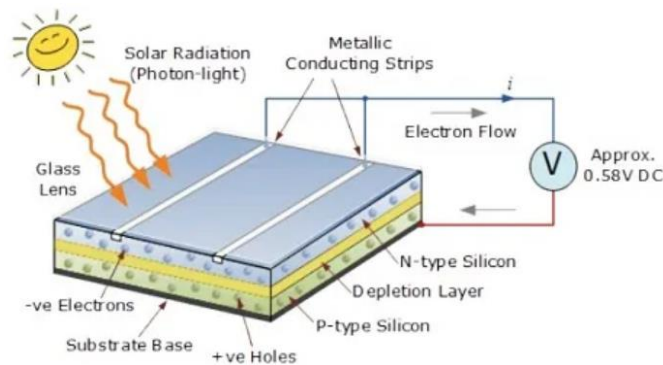


Fig.3: Working of solar cell

It is well known that doped semiconductors are of two types, p and n- types semiconductors depending upon the nature of the charge carriers. In n-type semiconductor the free carriers are electrons and in p-type semiconductor, the positive charge carriers are holes.

Since the semiconductors are electrically neutral, in a doped semiconductor the number of free carriers is equal to the lattice ions present in the semiconductor. A solar cell is illuminated by light having photon energy greater than the band gap energy of the solar cell. Then, using a proper circuit, the open circuit voltage, short circuit current and power drawn from the solar cell are measured.

Procedure:

1. Complete the circuit as shown in circuit diagram
2. Illuminate the solar cell. Adjust the R_h position for resistance so that the volt meter reads zero & ammeter reads a value of max value. This is the short circuit connection. Note down the value of the current as short circuited current, I_{sc}
3. Increase the resistance by varying the R_h slowly and note down the readings of current and voltage till a maximum voltage is read. Ensure to take at least 5 – 10 readings in this region
4. Disconnect the R_h and note down the voltage. This is the open circuit voltage, V_{oc} .
5. Repeat the experiment for another intensity of the illumination source
6. Tabulate all readings in Table 1. Calculate the power using the relation, $P = V \times I$
7. Plot I vs. V with I_{sc} on the current axis at the zero volt position and V_{oc} on the voltage axis at the zero current
8. Identify the maximum power point P_m on each plot. Calculate the series resistance of the solar cell using the formula as follows : $R_s = [\Delta V / \Delta I]$.
9. To see the performance of the cell calculate fill factor (FT) of the cell, which can be expressed by the formula, $FF = [P_m / I_{sc} V_{oc}]$.
10. Repeat the experiment with distance varying from source to solar cell for example say 50 cms
11. Note the reading of Voltmeter , Ammeter & also R_h Load resistance value by varying provided on Board. Tabulate all readings in Table 2

Observations and Table:

Source:1 _____			
S. No.	Voltage V (V)	Current I (A)	Power (VI) (watt)
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			
11			
12			
13			
14			
15			
16			
17			
18			
19			
20			

Source:2 _____			
S. No.	Voltage V (V)	Current I (A)	Power (VI) (watt)
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			
11			
12			
13			
14			
15			
16			
17			
18			
19			
20			

Calculations:

For source-1: maximum power point $P_m =$

Open circuit voltage $V_{oc} =$

Short circuited current $I_{sc} =$

Fill factor of the cell, $FF = \frac{P_m}{I_{sc}V_{oc}} =$

For source-2: maximum power point $P_m =$

Open circuit voltage $V_{oc} =$

Short circuited current $I_{sc} =$

Fill factor of the cell, $FF = \frac{P_m}{I_{sc}V_{oc}} =$

Change of maximum voltage $\Delta V =$

Change of maximum current with source $\Delta I =$

Series resistance of the solar cell: $R_s = [\Delta V / \Delta I] =$

Safety measures:

- Don't touch the solar panel (cell) with the light source.
- Set right position of rheostat for resistance so that the volt meter reads zero & ammeter reads a value of max value in short circuit connection.

Result: The V-I characteristics of solar cell have been studied.

For the given Solar cell, Series Resistance $R_s =$ ----- ohm;

and

Fill factor (FF) =----- for source-1

Fill factor (FF) =----- for source-2

Questions:

1. What is photo electric effect?
2. How is photo diode connected in the circuit? Forward or reverse? Why?
3. What is purpose of variable resistor?
4. What is change in voltage and current by changing light source?
5. What type of current is produced by solar cell either AC or DC?
6. What is difference between photo-voltaic cell and photo-conductive cell?
7. Is photo-current either wavelength dependent or intensity dependent or both? Justify.

DATE:

Roll No.:

9. STUDY THE B-H CURVE OF MAGNETIC MATERIAL

Objectives:

- To draw the curve between the magnetizing field (H) & the intensity of magnetization (I) of the specimen.
- If magnetizing field is H, intensity of magnetization is I then the magnetic induction (B) is given as $B = \mu_0(H + I)$

Learning Outcomes: After completing this experiment student is able to

- Realise the magnetization
- Understand the relation between magnetizing field (H) & the intensity of magnetization (I)
- Graphical analysis of magnetizing field (H) & the intensity of magnetization (I)
- Determine magnetic induction (B)

Resources required: Deflection Magnetometer (DM), A wooden bench with sliding magnetizing coil (solenoid) and a Compensating Coil, a Battery eliminator, a Rheostat, a Commutator & Plug key.

Key Words: Hysteresis, magnetizing field, intensity of magnetization, magnetic induction.

Principle: Solenoid is a type of electromagnet formed by a helical coil of wire whose length is substantially greater than its diameter, which generates a controlled magnetic field. The coil can produce a uniform magnetic field in a volume of space when an electric current is passed through it.

Formula: Magnetic field intensity, $H = \frac{4\pi n I}{10} = K_1 I$ C Amp/meter.

Where $K_1 = \frac{4\pi n}{10}$

n = No. of turns / cm. In the solenoid = _____ turns/cm.

The intensity of magnetization, defined as the magnetic moments per unit volume is given by:

$$I = \frac{H_e (d^2 - l^2) \tan \theta}{2d \cdot 2l \cdot a} \quad \text{amp/meter m}^2$$
$$= K_2 \tan \theta$$

Where $K_2 = \frac{H_e (d^2 - l^2)}{2d \cdot 2l \cdot a}$

a = Area of cross-section of the specimen = _____ cm^2

L = 5/12 of the geometric length of the specimen = _____ cm

d = distance between center of solenoid, and center of DM = _____ cm

H_e = Horizontal component of the earth's magnetic field
= 0.38 oersteds.

$K_1 =$ _____

$K_2 =$ _____

Circuit diagram:

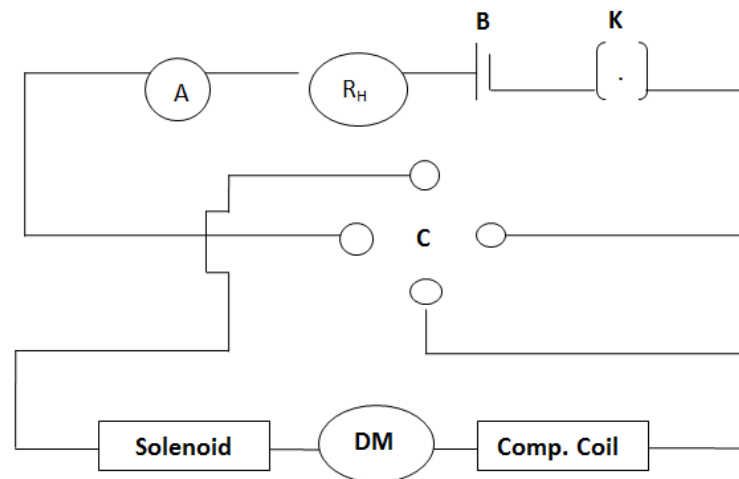


Fig. 1: Circuit diagram for magnetization and demagnetization

Model graph:

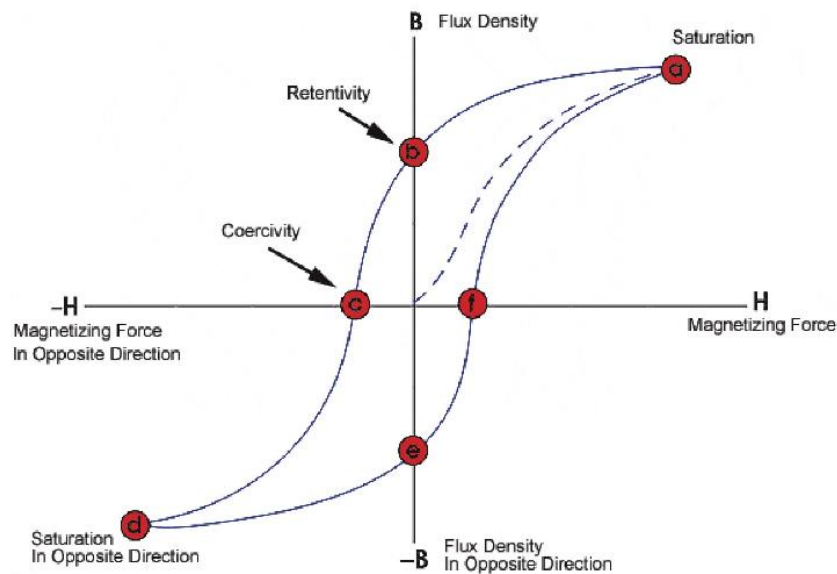


Fig.2: Plot a graph between C and I or between C and $\tan \theta$ or B and H .

Procedure:

1. Remove all magnets /magnetizing materials from the vicinity.
2. Draw the magnetic North-South direction on the table with a chalk piece using the Deflection Magnetometer (DM). Draw also the East-West Line.
3. Arrange the magnetometer scale along the East-West line. Set the aluminium pointer to read 0-0.
4. Make the circuit connections as shown. Get it checked by the instructor.
5. Without placing the specimen in the solenoid, close the circuit increase the current up to 2.5 amps in steps of 0.25 amps. Also adjust the compensating coil such that the deflection in DM is 0-0 (readings at the ends of the pointer) for all values of the current.
6. Placing the unmagnetized specimen in the solenoid with on the power supply and with minimum current note the readings at the two ends of the pointer. While doing so cover the image of the pointer by the pointer itself to avoid parallax error. Note the readings of the pointer ends it read 0-0 when the current (C) is zero. Increase the current by using Rheostats in 0.25 amps steps till the maximum current is reached and notes the corresponding reading in DM.

7. Now decrease the current in steps of 0.25 amps till you reach the minimum current, noting down the reading at every stage. For 0 amps reading, remove the key from the plug key.
8. Rotate the Commutator back of 90° . THIS WILL REVERSE THE DIRECTION OF THE CURRENT FLOW IN THE SOLENOID AND THE COMPENSATING COIL.
9. As before, Increase the current in steps of 0.01Amps till the deflections come to 0-0 in DM and afterwards increase the current in steps of 0.25 amps and read the ends of the pointer at each stage. When the maximum current is reached, decrease the current in 0.25Amps steps till you reach minimum current and note the readings. Remove the Key from the plug key when 0 Amps reading obtained
10. Again, rotate the Commutator back by 90° .
11. Increase the current in steps of 0.01Amps till we get 0-0 deflections in DM. Thereafter increase the current in steps of 0.25 amps and read the pointer ends. Reach the maximum current and note the readings REMOVE THE KEY FROM THE PLUG KEY. SWITCH OFF THE POWER SUPPLY.

OBSERVATIONS:

a = Area of cross-section of the specimen = _____ cm^2

L = $5/12$ of the geometric length of the specimen = _____ cm

d = distance between center of solenoid, and center of DM = _____ cm

H_e = Horizontal component of the earth's magnetic field

= 0.38 oersteds.

K_1 = _____

K_2 = _____

Table:

One Direction: By picking plugs in commutator

S.No	Current(C) Amperes	Deflection θ			$\tan\theta$
		θ_1	θ_2	mean θ	

S.No	Current(C) Amperes	Deflection θ			$\tan\theta$
		θ_1	θ_2	mean θ	

Opposite Direction: by changing plugs in commutator

S.No	Current(C) Amperes	Deflection θ			$\tan\theta$
		θ_1 mean θ	θ_2		

S.No	Current(C) Amperes	Deflection θ			$\tan\theta$
		θ_1 mean θ	θ_2		

S.No	Current(C) Amperes	Deflection θ			$\tan\theta$
		θ_1 mean θ	θ_2		

Safety measures:

- Set right position of rheostat for adjusting desired value of current flow in the solenoid.
- Don't keep other magnetic materials, magnets, current carrying coils, etc. in surroundings of the experiment.

Result:

The hysteresis behaviour of a magnetic solenoid is studied. The B-H curve is plotted.

Questions:

1. What is hysteresis?
2. Why magnetic induction is lagging with applied magnetic field?
3. What is purpose of variable resistor?
4. What is purpose of commutator?
5. What is a solenoid?
6. List applications of magnetic induction in daily life?
7. How do you distinguish soft and hard magnetic materials with the help of B-H curve?
8. What is the significance of area enclosed in the B-H curve?

DATE:

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10. DETERMINATION OF PLANCK'S CONSTANT

Objectives:

- Experimental determination of Planck's constant, $h = 6.626 \times 10^{-34} \text{ J} \cdot \text{s}$ using photocell.

Learning Outcomes: After completion of the experiment the student is able to

- Understand the concept of Photoelectric effect
- Realize the importance of Planck's constant
- Demonstrate Photo electric effect
- Calculate Photo current and Stopping Potential for various filters.

Resources Required: Photocell, Power supply, Micro ammeter, voltmeter, source of light and connecting wires.

Key Words: Photo electric effect, Incident radiation, frequency, Photo metal, Stopping Potential, Work Function.

Principle: The Photoelectric effect refers to the emission or ejection of electrons from the surface of a metal when light of sufficient frequency is incident on it.

Formula: During the interaction of the incident photon with the surface electron, the energy of the photon is completely absorbed by the electron and is used in the following two ways:

- (i) A part of the photon's energy is used up in coming out of the metallic surface
- (ii) Rest of the energy is gained by the electron as kinetic energy
If $h\nu = h\nu_0 + \frac{1}{2}mv^2$

Where $h\nu_0$ is the work function, which is constant for a particular emitter, If

eV_0 is the energy in electron volts,
 $eV_0 = h\nu - h\nu_0$

Where V_0 is the stopping potential applied to the anode.

$$\Rightarrow V_0 = \frac{h}{e}\nu - \frac{h}{e}\nu_0 \quad (3)$$

Thus, Plank's constant may be obtained from the slope of Stopping potential Vs frequency curve.

For particular colour, $eV_{01} = h\nu_1 - h\nu_0$ (4),
and for another colour, $eV_{02} = h\nu_2 - h\nu_0$ (5)

By Solving the above equations,

$$eV_{02} - eV_{01} = h\nu_2 - h\nu_1,$$

$$e(V_{02} - V_{01}) = h(\vartheta_2 - \vartheta_1),$$

$$h = \frac{e(V_{02} - V_{01})}{(\vartheta_2 - \vartheta_1)},$$

$$h = e * \text{slope},$$

$$\text{Slope} = \frac{dV}{d\vartheta} = \frac{\Delta V}{\Delta \vartheta} = \frac{(V_{02} - V_{01})}{(\vartheta_2 - \vartheta_1)}$$

Circuit Diagram:

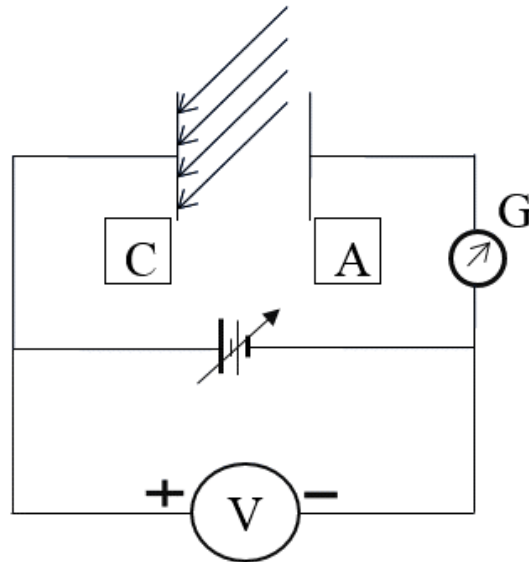


Fig.1: Photo-electric effect

Model graph: Stopping potential on the Y-axis Vs frequency on the X-axis.

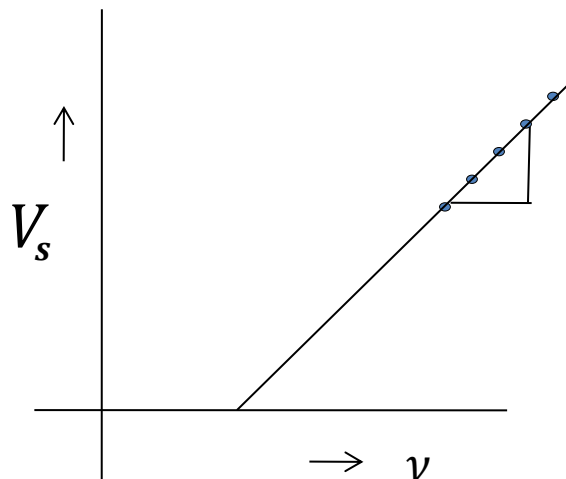


Fig.2: Stopping potential Vs Frequency

From graph, $h = e \frac{(V_{02} - V_{01})}{(\vartheta_2 - \vartheta_1)} = e \times \text{slope} =$

Js.

Theory: A Photoelectric cell constitutes two photo sensitive metals enclosed in an evacuated quartz bulb. The emission of electrons from a metallic plate when illuminated by light or electromagnetic radiation of suitable wavelength is called **Photoelectric Effect**. The plate used for the emission of electrons is called photo metal and the ejected electrons is called photo electrons.

The number of electrons emitted, and their kinetic energy depends on the

- (a) The potential difference between the two electrodes
- (b) The intensity of incident radiation
- (c) The frequency of incident radiation and
- (d) The photo metal used

For a given photo metal, keeping the intensity and frequency of incident radiation fixed, consider the effect of potential difference between the plates. The anode potential can be made positive or negative with respect to the cathode. As the positive potential of the anode is increased, the electrons which are emitted get accelerated towards anode and contribute to the current flowing in the circuit. Thus, the photo current is increased. However, if the positive potential is further increased, the current remains constant. As all the photo electrons emitted would have already reached the plate, further increase in potential would mean no more electrons available to reach anode.

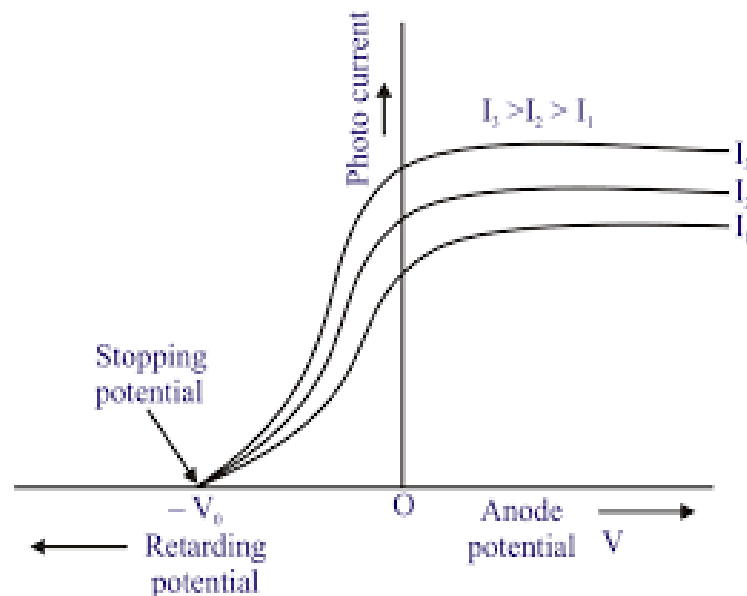


Fig.3: Photocurrent Vs Anode potential

If the potential difference is made negative, the photo current does not immediately drop to zero, but continues to flow in the same direction. This shows that photo electrons are emitted from the plate with a finite velocity. If the negative or retarding potential is further increased, the photo current decreases and finally becomes zero. The negative potential on the plate at which the photo current becomes zero is called as **stopping potential**. It is the potential difference between the two electrodes which is just sufficient to halt the most energetic electrons emitted.

Procedure:

- (1) The Photocell unit is taken and a bulb of 40 to 60 watts is fixed.
- (2) A suitable filter of known wavelength is inserted in front of the photocell with the power supply switch towards the negative anode potential mode and the voltmeter in the (0-1) V range.
- (3) The photocell unit is switched ON with bulb OFF and the Supply knob adjusted to zero applied voltage. Note that both the voltmeter and the micro-ammeter indicate zero.
- (4) Now the bulb is switched ON to note the micro ammeter reading. This is 'zero voltage deflection'.
- (5) Vary the supply knob to apply negative voltage to the anode until the micro ammeter reading reduces to zero.
- (6) Note the voltmeter reading which indicates the 'Stopping Potential'.
- (7) Repeat the procedure with different filters and note the respective stopping potentials.
- (8) A graph is plotted between the stopping potential and the frequency of the filter.

Observations: $C = \text{Speed of light} = 3 \times 10^8 \text{ m/s.}$ and $e = 1.6 \times 10^{-19} \text{ C}$

Standard value of Planck's constant= $6.625 \times 10^{-34} \text{ Joule-Sec}$

S.No	Colour	Wavelength (λ) ° A	Frequency ($\nu=C/\lambda$) Hz	Stopping Potential V_s (volt)
1	red	6910	4.3×10^{14}	
2	yellow	5780	5.1×10^{14}	
3	green	5460	5.4×10^{14}	
4	blue	4360	6.8×10^{14}	

Calculations:

$$\text{Stopping potential } h = e \frac{(V_{02} - V_{01})}{(\nu_2 - \nu_1)} = \quad \text{Js}$$

Percentage Error: $\frac{(\text{Experimental Value} - \text{Standard Value})}{\text{Standard Value}} \times 100 = \text{-----}\%$

Safety Measures:

- 1) The observations should be taken carefully by alternating anode potential.
- 2) While performing experiment avoid stray light on photocell.
- 3) Stopping potentials should be noted down carefully.

Result: 1) Observed Value of Planck 's constant: ----- Joule-Sec

2) Percentage Error -----%

Questions:

1. What is Photo electric effect?
2. Define Stopping Potential.
3. What is a Photocell?
4. What is the purpose of using photocell in this experiment?
5. What are the types of Photo-electric cells?
6. Explain Einstein Photoelectric equation.
7. Are all the metals useful for Photoelectric effect?
8. What is the significance of Photoelectric effect?
9. Define Work Function.
10. Define Electron Volt.
11. Can we find ionization potential of a given metal by this experiment? How?
12. What is the range of Photo current during your experiment?
13. What is the value of Photo current at stopping Potential? Why?
14. How can you justify, Photoelectric effect supports quantum nature of light?
15. It is found that green light does not eject photoelectrons from a metal. Is it advisable to try with blue light? With red light?
16. A non-monochromatic light is used in an experiment on photoelectric effect. What could be the stopping potential?

DATE:

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11. DETERMINATION OF HALL COEFFICIENT AND CARRIER CONCENTRATION OF A GIVEN SEMICONDUCTOR

Objectives:

- To determine Hall co-efficient, the type of semiconductor and concentration of charge carriers in an extrinsic semiconductor by Hall Experiment.

Learning Outcomes: After completion of the experiment the student is able to

- Realize the effect of Lorentz force acting on charge carriers in a semiconductor.
- Distinguish that the Hall co-efficient is negative for n-type semiconductor and it is positive for p-type semiconductor.
- Understand the generation of magnetic flux through Electro-magnets.
- Utilize the Gauss meter for measurement of Magnetic Induction.

Resources Required: Commercial setup with the following components: electromagnet with power supply, Hall probe, Semiconductor sample, arrangement for pressure contact, current supply with meter, voltmeter etc.

Key Words: Hall Effect, Hall voltage, Hall co-efficient, density of charge carriers, n-type semiconductor, p-semiconductor and Magnetic induction.

Principle: When piece of current carrying semiconductor be placed in a transverse magnetic field then there induces electric field mutually perpendicular to both the current flow and the applied magnetic field. This phenomenon is called Hall Effect.

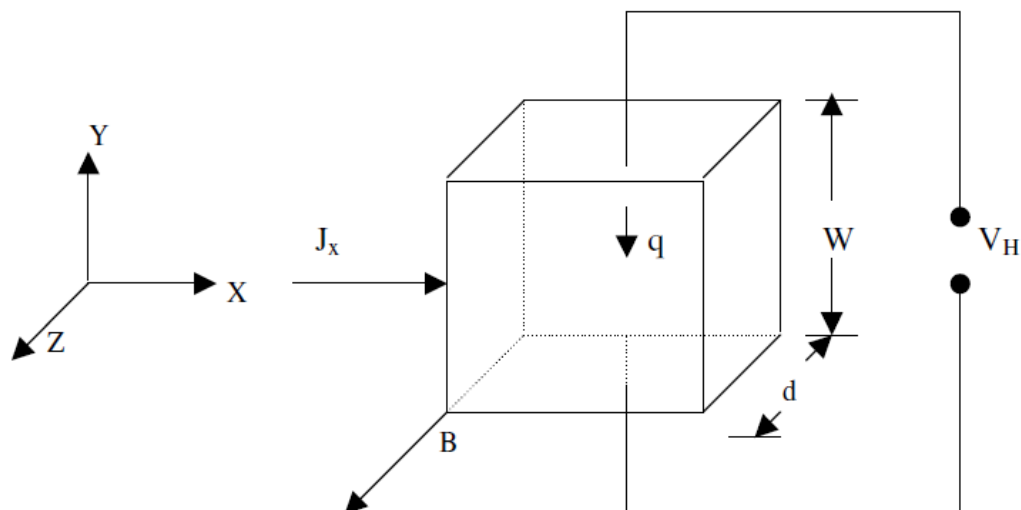


Fig. 1: Hall Effect

Formula: The Hall coefficient, $R_H = \frac{V_H d}{IB}$ ----- (i) (from Table)

where current (I), Hall voltage (V_H), thickness (d) of the semiconductor and applied magnetic field (B).

and $R_H = \frac{d(m)}{B}$ ----- (ii) (from Graph)

where, m= slope of a straight line V_H versus I.

Concentration of charge carriers is given by

$n = -\frac{1}{R_H e}$ ----- (iii) for n-type semiconductor.

$p = +\frac{1}{R_H e}$ ----- (iv) for p-type semiconductor.

where e is charge of electron.

Experimental set-up:

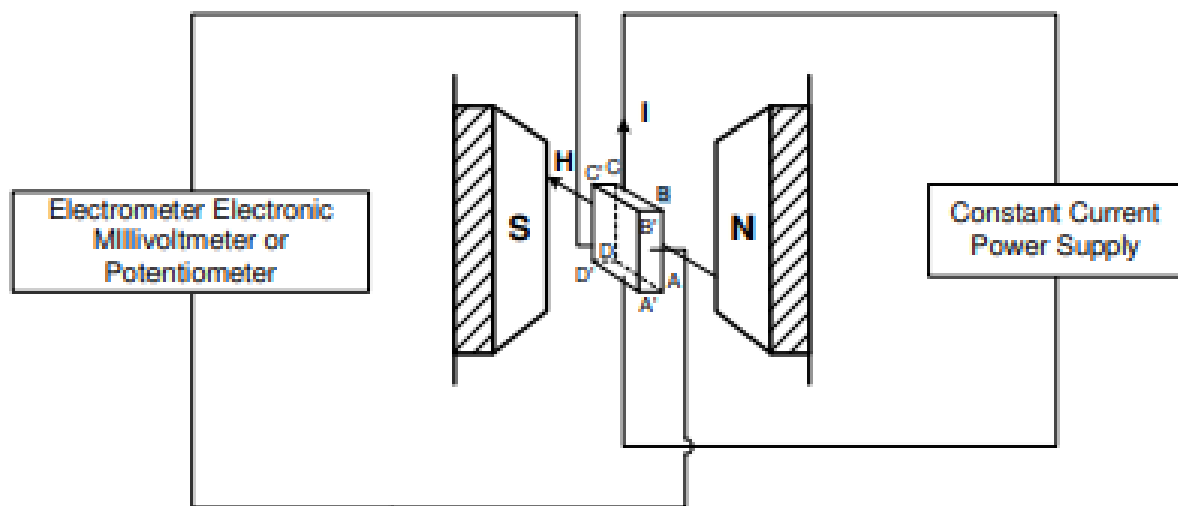


Fig. 2: Hall Experiment

Model Graph:

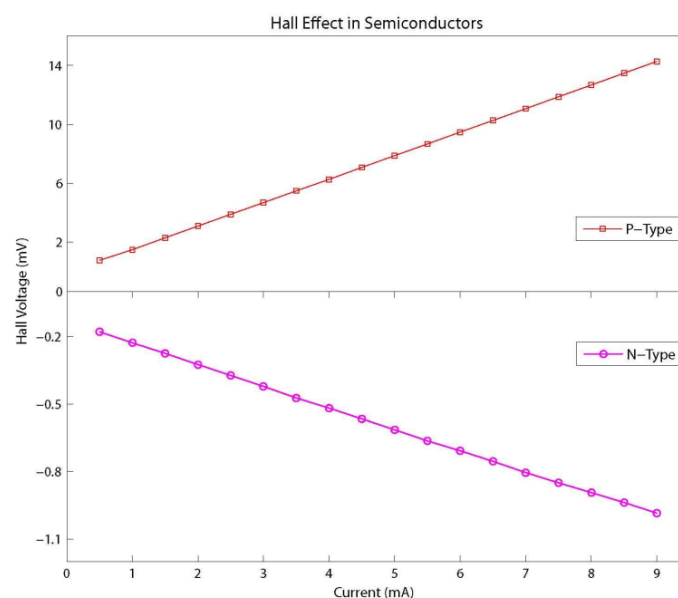


Fig. 3: Voltage as a function of Current

Theory: Consider a rectangular slab of semiconductor with thickness d kept in XY plane [see Fig. 1(a)]. An electric field is applied in x-direction so that a current (I) flows through the sample. If w is width of the sample and d is the thickness, the current density is given by $J_x = I/wd$. Now a magnetic field B is applied along positive z axis (Fig. 1). If the charge carriers are positive (negative) and are moving with velocity v along positive (negative) x-axis then the direction of force experienced by the charge carriers in presence of magnetic field is along negative y direction. This results in accumulation of charge carriers towards bottom edge (Fig. 1). This sets up a transverse electric field E_y in the sample. The potential, thus developed, along y-axis is known as Hall voltage V_H and this effect is called Hall Effect.

Assuming E_y to be uniform the Hall voltage is given by $V_H = E_y w$ ----- (1)

and the Hall coefficient R_H is given by $R_H = \frac{E_y d}{J_x B} = \frac{V_H d}{IB}$ ----- (2)

The carrier density is related to the Hall coefficient by the relation

$$n = -\frac{1}{R_H e} \text{ ----- (3) for n-type semiconductor.}$$

$$p = +\frac{1}{R_H e} \text{ ----- (4) for p-type semiconductor.}$$

where e is charge of electron.

PROCEDURE:

1. Connect the leads from the sample to the "Hall effect Set-up" unit. Connect the electromagnet to constant current generator.
2. Switch on the current through sample and measure the hall voltage without any magnetic field. There may be some voltage due to misalignment of pressure contacts on the sample. This error must be subtracted from the readings.
3. Switch on the electromagnet and set suitable magnetic field (<3 K gauss). We can measure this using Hall probe. (Set magnetic field $B=2$ kG and $B=3$ kG for the experiment).
4. Insert the sample between the pole pieces of the electromagnet such that I , B and V are in proper direction (Fig.1).
5. Record the Hall voltage and also record voltage by reversing both the current and magnetic field simultaneously. (Note down data for the first two columns with $+B$ for all I 's and then reverse the field ($-B$) to record data for the next two columns)
6. Keeping the magnitude of magnetic field constant, measure Hall voltage as a function of I .
7. Repeat step 5 and 6 for various magnetic fields.
8. Plot V_H as a function of I using the averaged data and find the value of Hall coefficient from the slope of the graph. Hence determine charge carrier density and type of majority carrier in the given material.
9. Note down the type of the sample.

Observations:

Thickness of the sample (d)=----- (m)

Applied Magnetic field (B)= ----- Gauss

Sample type: -----

Calculations:

Slope of the straight line (m) = ----- V/C

Hall coefficient R_H is given by $R_H = \frac{d(m)}{B} = \text{----- } m^3/C$ (from Graph)

The carrier density is given by $n = \frac{1}{R_H e} =$

TABLE: 1 Voltage as a function of Current

S.No	I (A)	V_H (V)	$R_H = \frac{V_H d}{IB} \text{ m}^3/C$
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			

Average $R_H = \text{----- } m^3/C$ (From Table)

Safety measures:

1. The equipment must remain Switched off for at least fifteen minutes before being switched on again.
2. Do not operate the Instrument around explosive gas, vapour or dust.

Results: The Hall Effect has been studied.

1. Type of semiconductor is -----
2. Hall co-efficient $R_H =$ ----- m^3/C (From Table)
3. Hall co-efficient $R_H =$ ----- m^3/C (From Graph)

Questions:

1. What is Hall Effect?
2. What are n-type and p-type semiconductors?
3. What is the effect of temperature on Hall coefficient of a lightly doped semiconductor?
4. Do the holes actually move?
5. Obtain the Carrier density in a given semi- conductor from the Hall coefficient.
6. Why the resistance of the sample increases with the increase of magnetic field?
7. Why a high input impedance device is generally needed to measure the Hall voltage?

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12. MEASUREMENT OF DIELECTRIC CONSTANT

Objective: To Determine the dielectric constant (K) of a given material by resonance method.

Learning Outcomes: After completion of this experiment, the student is able to

- Understand the working of dielectric material.
- The working of different dielectric materials when inserted between capacitor plates.
- Realize the importance of dielectric materials

Resources required: Dielectric constant apparatus, variable gang capacitor, test capacitor, dielectric material, connecting patch cords.

Keywords: Dielectric constant, dielectric material, variable capacitor, resonance.

Principle: *The gap between the plates filled completely by dielectric material in a capacitor will increase its capacitance by the factor of the dielectric constant value.*

Formula: Dielectric constant, K is given by

$$K = \frac{C_1 - C_2}{C_1 - C_3}$$

Where

C_1 = Capacity of standard variable capacitor at resonance (maximum deflection)

C_2 = Capacity of standard variable capacitor at resonance (maximum deflection) including test capacitor with dielectric in it

C_3 = Capacity of standard variable capacitor at resonance (maximum deflection) including test capacitor without dielectric in it

Experimental Set-up:

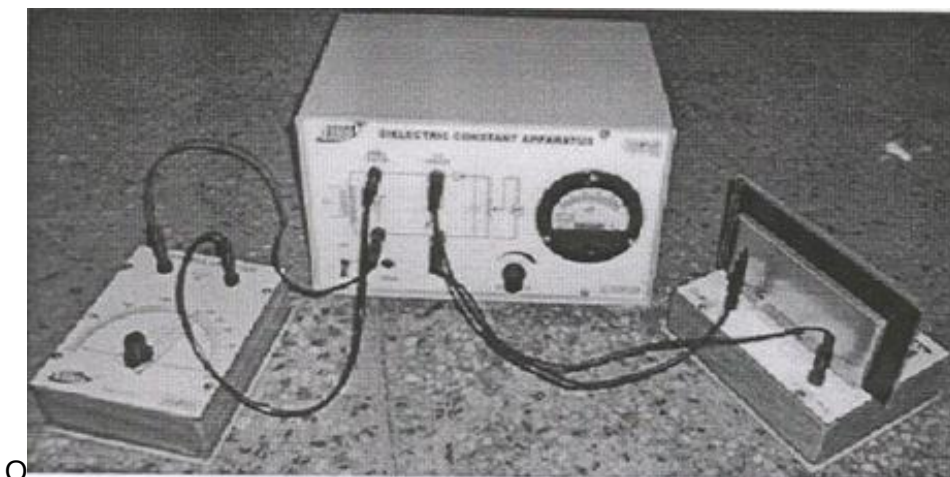


Fig. 1: Set up to determine Dielectric constant.

Theory: A capacitor is an electronic component designed to store electric charge. This is widely built by sandwiching a dielectric insulating plate in between the metal conducting plates. The dielectric property plays a major role in the functioning of a capacitor.

The layer made up of dielectric material decides, how effectively the capacitor can store the charge. Picking the right dielectric material is crucial. Thus, we can also define it as 'the ratio of the electric field without a dielectric(E_0) to the net field with a dielectric(E).'

$$K = \frac{E}{E_0}$$

Here, the value of E_0 is always greater than or equal to E . Thus, the value of a dielectric constant is always greater than 1.

The greater the value of K the more charge can be stored in a capacitor.

In the capacitor, the capacitance is given by $C = KC_0$

Thus, filling the gap between the plates completely by dielectric material will increase its capacitance by the factor of the dielectric constant value.

In the parallel plate capacitor, the capacitance is given by:

$$C = K\epsilon_0 \frac{Ad}{d}$$

Where,

- C is the capacitance of the parallel plate capacitor.
- K is the dielectric constant.
- ϵ_0 is the permittivity of the free space.
- A is the area of parallel conducting plates
- d is the separation between parallel conducting plates

The capacitance value can be maximized by increasing the value of the dielectric constant and by decreasing the separation between the parallel conducting plates.

Dielectric Constant Value: The value of the dielectric constant is crucial in building various electronic components. The following table gives some typical values of dielectric constants:

Table: 1 Dielectric constant values.

Dielectric Materials	Dielectric Constant Value
The dielectric constant of vacuum	1.00
The dielectric constant of air	1.00059
The dielectric constant of water	80
The dielectric constant of paper	3.6

Factors affecting Dielectric Constant

The dielectric constant depends on various factors such as:

- 1) **Frequency:** The frequency of the applied voltage is one of the factors affecting dielectric constant. As the frequency of the applied voltage increases, the value of the dielectric constant becomes non-linear.

- 2) **Applied voltage:** When a direct current voltage is applied, the value of the dielectric constant reduces while applying alternating current voltage would increase the value of the dielectric constant.
- 3) **Temperature:** When the temperature is low, the alignment of the molecules in the dielectric material is difficult. By increasing the temperature, the dipoles in the dielectric material become dominant resulting in an increase in the dielectric constant. This temperature is known as the transition temperature. If the temperature rises above the transition temperature, then there will be a gradual decrease in the dielectric constant.
- 4) **Humidity and moisture:** The strength of the dielectric material decreases when either the humidity or the moisture is increased.
- 5) **Heating effect:** When the dielectric material is heated, the dielectric loss takes place. Dielectric loss is defined as the dissipation of energy in the form of heat when there is a movement of the molecules in the material, as it is exposed to the alternating current voltage. This takes place as the material absorbs electrical energy.
- 6) The structure and morphology of the material also influence the dielectric constant.
- 7) Deterioration and weathering of the material also affect the dielectric constant.

Procedure:

1. Connect the variable capacitor with the terminal marked variable capacitor on the front panel of the apparatus
2. Switch ON the instruments and then set the sensitivity by the sensitivity control knob and also rotate the standard capacitor knob for maximum deflection. At one position of standard variable capacitor the deflection should be near 85 to 90uA
3. Now vary the variable capacitor to find out the resonance point (maximum deflection in the meter). Note the value of variable capacitor as C_1 .
4. After getting the value of C_2 , do not disturb the sensitivity knob for whole the experiment.
5. Now include the unknown test capacitor (with dielectric(all the three Bakelite sheets - 7.5mm in it) and repeat the experiment by varying variable capacitor to again obtain resonance point (maximum deflection) in the meter. Note this value of variable capacitor as C_2
6. Now remove the dielectric material (Bakelite sheets) from the test capacitor and repeat the experiment by varying variable capacitor to again obtain resonance point (maximum deflection) in the meter. Note this value of variable capacitor as C_3
7. Repeat the whole experiment with different thickness of materials
8. Calculate the value of dielectric constant K using above formula.

Observations:

1. C_1 = Capacity of standard variable capacitor at resonance (maximum deflection)= _____pf
2. C_2 = Capacity of standard variable capacitor at resonance (maximum deflection) including test capacitor with dielectric in it= _____pf
3. C_3 = Capacity of standard variable capacitor at resonance (maximum deflection) including test capacitor without dielectric in it= _____pf

Table: 2 To determine Dielectric Constant for different materials

S.No	Thickness of dielectric material (Bakelite sheets) in mm	Alone C_1 (pf)	With test capacitor (with dielectric sample in it) C_2 (pf)	With test capacitor (without dielectric sample in it) C_3 (pf)	Dielectric constant (K)
1	3mm				
2	4.5mm				
3	6mm				
4	7.5mm				

Precautions:

1. Give the circuit connections properly.
2. Test Capacitor plates should be tight when filled with dielectric material
3. Three sheets are provided are with one 1.5mm thick and two 3mm each thick, by using all three sheets the thickness will be 7.5mm

Result: The dielectric constant for different sheets is as follows:

S.No.	Type of the dielectric sheet	dielectric constant
1		
2		
3		
4		

Questions:

1. What are dielectrics? Give examples.
2. What is a capacitor?
3. Define capacitance of a capacitor?
4. Mention the formula for capacitance of a capacitor.
5. How do you increase the capacitance of a capacitor?
6. Define dielectric constant?
7. What is dielectric strength?
8. Mention the SI unit of dielectric constant.
9. What happens when dielectric is introduced between the plates of a capacitor?
10. Why dielectric constant is different for different materials.
11. Why do we use high resistance in the circuit?
12. What is dielectric loss?
13. What is effect of electric field, potential on capacitance of a capacitor?
14. Guess the possible reason for the water to have much larger dielectric constant?
15. What do you mean by dielectric polarization?
16. Mention types of capacitors based on dielectrics used.
